

Supplementary Information

A Coefficient- and Phase-Resolved Bridge across Noncommuting Limits of Lamb-Wave ZGV Splitting

Contents

1	Exact isotropic Rayleigh–Lamb branch and its Morse–Bott ring	1
1.1	Helmholtz potentials and symmetric parity	2
1.2	Face tractions and the original determinant	2
1.3	Desingularization and the shear cutoff	3
1.4	Implicit dispersion derivatives	4
1.5	Cartesian Hessian and the Morse–Bott ring	4
2	Weak-anisotropy normal form and Morse splitting	6
2.1	General Hermitian-pencil sensitivity	6
2.2	Mass gauge, augmented solve, and the complete radial coefficient	6
2.3	Rotation and complete contraction of the cubic perturbation	7
2.4	Controlled normal form and radial implicit-function reduction	9
2.5	Exact Cartesian Hessian and its Schur inertia	10
2.6	General even harmonic and the cubic eight-point theorem	10
2.7	Exceptional cases and theorem boundaries	11
2.8	Conversion to the dimensionless computational convention	12
3	Bessel–Morse asymptotics of the Green response	12
3.1	Fourier convention and normalization-invariant branch projection	12
3.2	Local polar integral and radial stationary phase	13
3.3	The joint transition limit and its exact Bessel kernel	14
3.4	General angular weights and the Fourier–Bessel selection rule	15
3.5	Ring, overlap, and fixed-anisotropy regimes	15
3.6	Exact Cartesian Morse theorem at fixed anisotropy	17
3.7	Frequency separation and signed modulation	18
3.8	Nodal channels, noncommuting limits, and theorem boundaries	18
4	Thickness-resolved spectral numerics and error control	18
4.1	GLL element map and traction-free weak form	19
4.2	Matrix derivatives and eigensystem diagnostics	19
4.3	Sensitivities and critical-point resolution checks	20
4.4	Polar Green quadrature, fit masks, and phase accuracy	22
4.5	Declared configuration and fixed acceptance thresholds	25
4.6	Qualified numerical-artifact registry	26
4.7	Reconstruction boundary	29
5	Generated convergence records and robustness figures	30

1 Exact isotropic Rayleigh–Lamb branch and its Morse–Bott ring

We derive the symmetric Rayleigh–Lamb equation from the displacement potentials, retain the physical traction matrix before any algebraic division, and then construct the determinant used by the numerical solver. The plate occupies

$\{(x, y, z) : -h \leq z \leq h\}$ and is infinite in the in-plane coordinates. Its density is $\rho > 0$, and its isotropic stiffness is specified by the Lamé moduli λ and μ , with

$$c_L^2 = \frac{\lambda + 2\mu}{\rho}, \quad c_T^2 = \frac{\mu}{\rho}. \quad (1)$$

We use the time convention $\exp[-i\omega t]$. Isotropy lets us rotate any in-plane wave vector onto the x axis while deriving the scalar equation; the two-dimensional wave-vector dependence is restored at the end.

1.1 Helmholtz potentials and symmetric parity

In the sagittal (x, z) plane, the displacement $\mathbf{u} = (u_x, 0, u_z)$ obeys Navier's equation

$$\rho \partial_t^2 \mathbf{u} = (\lambda + \mu) \nabla(\nabla \cdot \mathbf{u}) + \mu \nabla^2 \mathbf{u}. \quad (2)$$

Introduce a longitudinal potential ϕ and a sagittal shear potential ψ by

$$\mathbf{u} = \nabla \phi + \nabla \times (\psi \mathbf{e}_y), \quad u_x = \partial_x \phi - \partial_z \psi, \quad u_z = \partial_z \phi + \partial_x \psi. \quad (3)$$

The divergence of the curl term is zero and the curl of the gradient term is zero. Taking the divergence and the sagittal curl of Eq. (2), respectively, gives the two scalar wave equations

$$(\nabla^2 - c_L^{-2} \partial_t^2) \phi = 0, \quad (\nabla^2 - c_T^{-2} \partial_t^2) \psi = 0. \quad (4)$$

For a factor $\exp[i(kx - \omega t)]$, define the longitudinal and shear thickness wavenumbers on their principal complex branches by

$$p^2 = \frac{\omega^2}{c_L^2} - k^2, \quad s^2 = \frac{\omega^2}{c_T^2} - k^2. \quad (5)$$

The symmetric Lamb parity is

$$u_x(x, -z) = u_x(x, z), \quad u_z(x, -z) = -u_z(x, z). \quad (6)$$

Equation (3) therefore requires an even longitudinal potential and an odd shear potential. With physical potential amplitudes A and B , the complete symmetric separated ansatz is

$$\phi = A \cos(pz) e^{i(kx - \omega t)}, \quad \psi = B \sin(sz) e^{i(kx - \omega t)}. \quad (7)$$

Suppressing the common exponential factor from now on, direct substitution gives

$$u_x(z) = ikA \cos(pz) - sB \cos(sz), \quad (8)$$

$$u_z(z) = -pA \sin(pz) + ikB \sin(sz). \quad (9)$$

The first expression is even and the second is odd, so the chosen potential parities reproduce Eq. (6) explicitly.

1.2 Face tractions and the original determinant

For an isotropic solid, $\boldsymbol{\sigma} = \lambda(\nabla \cdot \mathbf{u})\mathbf{I} + 2\mu\boldsymbol{\varepsilon}$, where $\boldsymbol{\varepsilon} = (\nabla \mathbf{u} + \nabla \mathbf{u}^\top)/2$. The derivatives of Eqs. (8) and (9) give

$$\varepsilon_{xx} = -k^2 A \cos(pz) - iksB \cos(sz), \quad (10)$$

$$\varepsilon_{zz} = -p^2 A \cos(pz) + iksB \cos(sz), \quad (11)$$

$$2\varepsilon_{xz} = -2ikpA \sin(pz) + (s^2 - k^2)B \sin(sz). \quad (12)$$

Consequently,

$$\frac{\sigma_{xz}}{\mu} = -2ikpA \sin(pz) + (s^2 - k^2)B \sin(sz). \quad (13)$$

For the normal stress, the shear-potential terms cancel from the dilatation, and hence

$$\sigma_{zz} = -[\lambda k^2 + (\lambda + 2\mu)p^2] A \cos(pz) + 2i\mu ksB \cos(sz). \quad (14)$$

$$\lambda k^2 + (\lambda + 2\mu)p^2 = \rho\omega^2 - 2\mu k^2 = \mu \left(\frac{\omega^2}{c_T^2} - 2k^2 \right) = \mu(s^2 - k^2), \quad (15)$$

$$\frac{\sigma_{zz}}{\mu} = -(s^2 - k^2)A \cos(pz) + 2i\mu ksB \cos(sz). \quad (16)$$

Here Eq. (15) uses both relations in Eq. (5) and $\rho c_T^2 = \mu$.

Traction freedom means $\sigma_{xz} = \sigma_{zz} = 0$ on each face. At the upper face $z = +h$, Eqs. (13) and (16) become

$$\underbrace{\begin{pmatrix} -2ikp \sin(ph) & (s^2 - k^2) \sin(sh) \\ -(s^2 - k^2) \cos(ph) & 2iks \cos(sh) \end{pmatrix}}_{T_S(k, \omega)} \begin{pmatrix} A \\ B \end{pmatrix} = \mathbf{0}. \quad (17)$$

At the lower face $z = -h$, oddness of σ_{xz} and evenness of σ_{zz} give the separate equations

$$\begin{pmatrix} 2ikp \sin(ph) & -(s^2 - k^2) \sin(sh) \\ -(s^2 - k^2) \cos(ph) & 2iks \cos(sh) \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = \mathbf{0}. \quad (18)$$

The first row of Eq. (18) is the negative of the first row of Eq. (17), while the second rows coincide. Therefore either face supplies the same two independent scalar conditions, and satisfaction on one face enforces satisfaction on the other for a mode of the stated parity.

Before any amplitude rescaling, the determinant of the physical traction matrix in Eq. (17) is

$$\begin{aligned} F_S(k, \omega) &= \det T_S \\ &= [-2ikp \sin(ph)] [2iks \cos(sh)] - [(s^2 - k^2) \sin(sh)] [-(s^2 - k^2) \cos(ph)] \\ &= 4k^2 p s \cos(sh) \sin(ph) + (s^2 - k^2)^2 \sin(sh) \cos(ph). \end{aligned} \quad (19)$$

Thus $F_S = 0$ is the usual symmetric Rayleigh–Lamb equation wherever the chosen potential basis is nonsingular.

1.3 Desingularization and the shear cutoff

At the shear cutoff $s = 0$, the basis function $B \sin(sz)$ collapses identically and both entries in the second column of T_S vanish. The resulting factor of s in Eq. (19) is therefore a basis normalization zero, not a physical dispersion condition. Define the regular shear amplitude

$$\hat{B} = sB, \quad B \sin(sz) = \hat{B} \frac{\sin(sz)}{s}. \quad (20)$$

For $s \neq 0$, the physical and regular amplitude vectors are related by $(A, B)^\top = \text{diag}(1, s^{-1})(A, \hat{B})^\top$. Hence the regular traction matrix is

$$\begin{aligned} \tilde{T}_S &= T_S \text{diag}(1, s^{-1}) \\ &= \begin{pmatrix} -2ikp \sin(ph) & (s^2 - k^2) \frac{\sin(sh)}{s} \\ -(s^2 - k^2) \cos(ph) & 2ik \cos(sh) \end{pmatrix}. \end{aligned} \quad (21)$$

Taking its determinant entry by entry gives

$$D_S(k, \omega) := \det \tilde{T}_S = (s^2 - k^2)^2 \frac{\sin(sh)}{s} \cos(ph) + 4k^2 p \cos(sh) \sin(ph). \quad (22)$$

For $s \neq 0$, Eqs. (19) and (22) satisfy exactly

$$F_S = sD_S. \quad (23)$$

They therefore have identical roots away from the shear cutoff. The regular matrix also defines the correct analytic continuation through that cutoff.

Indeed, the even entire function multiplying the second potential has the Taylor series

$$\frac{\sin(sh)}{s} = h - \frac{s^2 h^3}{3!} + \frac{s^4 h^5}{5!} - \frac{s^6 h^7}{7!} + O(s^8), \quad s \rightarrow 0. \quad (24)$$

Together with $\cos(sh) = 1 - s^2 h^2/2 + O(s^4)$, this gives the finite matrix and determinant limits

$$\tilde{T}_S|_{s=0} = \begin{pmatrix} -2ikp \sin(ph) & -k^2 h \\ k^2 \cos(ph) & 2ik \end{pmatrix}, \quad (25)$$

$$D_S|_{s=0} = k^4 h \cos(ph) + 4k^2 p \sin(ph). \quad (26)$$

At this cutoff $\omega = c_T k$ and $p^2 = (c_T^2/c_L^2 - 1)k^2$; Eq. (26) is not identically zero. It retains a zero only when the two regular face-traction equations are linearly dependent, which is the physical condition.

The other square root causes no loss of analyticity: $\cos(ph)$ and $p \sin(ph)$ are entire functions of p^2 , while $\cos(sh)$ and $\sin(sh)/s$ are entire functions of s^2 . Equation (5) therefore supplies analytic continuation into evanescent regions. For real k and ω , the continued determinant remains real even when p or s is purely imaginary.

1.4 Implicit dispersion derivatives

Let (k_0, ω_0) be a root of D_S with

$$D_{S,\omega}(k_0, \omega_0) \neq 0. \quad (27)$$

The implicit-function theorem then supplies a unique smooth local branch $\omega = \omega(k)$ satisfying

$$D_S(k, \omega(k)) = 0. \quad (28)$$

In the following equations, every partial derivative of D_S is evaluated at $(k, \omega(k))$. One total derivative of Eq. (28) gives

$$0 = \frac{d}{dk} D_S(k, \omega(k)) = D_{S,k} + D_{S,\omega} \omega_k, \quad (29)$$

$$\omega_k = -\frac{D_{S,k}}{D_{S,\omega}}. \quad (30)$$

Differentiating Eq. (29) once more, including the dependence of each partial derivative on both arguments, yields

$$\begin{aligned} 0 &= \frac{d}{dk} (D_{S,k} + D_{S,\omega} \omega_k) \\ &= (D_{S,kk} + D_{S,k\omega} \omega_k) + (D_{S,\omega k} + D_{S,\omega\omega} \omega_k) \omega_k + D_{S,\omega} \omega_{kk} \\ &= D_{S,kk} + 2D_{S,k\omega} \omega_k + D_{S,\omega\omega} \omega_k^2 + D_{S,\omega} \omega_{kk}, \end{aligned} \quad (31)$$

where equality of mixed derivatives was used in the last line. Solving this equation gives the general second derivative

$$\omega_{kk} = -\frac{D_{S,kk} + 2D_{S,k\omega} \omega_k + D_{S,\omega\omega} \omega_k^2}{D_{S,\omega}}. \quad (32)$$

A zero-group-velocity point satisfies $\omega_k(k_0) = 0$. Under Eq. (27), this is equivalent to $D_{S,k}(k_0, \omega_0) = 0$. Substitution into Eq. (32) leaves the radial curvature

$$a := \omega_{kk}(k_0) = -\left. \frac{D_{S,kk}}{D_{S,\omega}} \right|_{(k_0, \omega_0)}. \quad (33)$$

Thus, with $q = k - k_0$, Taylor's theorem gives

$$\omega(k) = \omega_0 + \frac{a}{2} q^2 + O(q^3). \quad (34)$$

For the selected first symmetric branch (the S_1 -ZGV branch within the configured search window) and the nondimensional reference values $h = \rho = \mu = 1$ and $\lambda = 2$, so that $c_T = 1$ and $c_L = 2$, the independent arbitrary-precision refinement of $D_S = D_{S,k} = 0$, seeded only by the float-valued branch search, is

$$k_0 = 0.8042173193715181, \quad \omega_0 = 2.8517587749600901, \quad a = 1.1968627250739301 > 0. \quad (35)$$

The executable check associated with this derivation differentiates Eq. (22) at arbitrary precision, verifies $D_S = D_{S,k} = 0$, verifies $D_{S,\omega} \neq 0$, and compares all three values in Eq. (35) with both the Task-4 solver and the generated reference artifact. This is a separate high-precision refinement of the same exact scalar determinant; the independent thickness-resolved spectral cross-check is supplied by the numerical validation rather than claimed here.

1.5 Cartesian Hessian and the Morse–Bott ring

Restore the two-dimensional wave vector $\mathbf{k} = (k_x, k_y)$, and write

$$r = |\mathbf{k}| = (k_x^2 + k_y^2)^{1/2}, \quad \mathbf{e}_r = \frac{\mathbf{k}}{r}, \quad \mathbf{e}_\theta = (-e_{r,y}, e_{r,x}). \quad (36)$$

For $r > 0$, isotropy makes the selected branch radial: $\omega^{(0)}(\mathbf{k}) = f(r)$, where $f(k) = \omega(k)$ in the one-dimensional derivation above. Since

$$\partial_{k_i} r = \frac{k_i}{r}, \quad \partial_{k_j} \left(\frac{k_i}{r} \right) = \frac{\delta_{ij}}{r} - \frac{k_i k_j}{r^3}, \quad (37)$$

the Cartesian gradient and Hessian follow component by component:

$$\partial_{k_i} \omega^{(0)} = f'(r) \frac{k_i}{r}, \quad (38)$$

$$\partial_{k_j} \partial_{k_i} \omega^{(0)} = f''(r) \frac{k_i k_j}{r^2} + f'(r) \left(\frac{\delta_{ij}}{r} - \frac{k_i k_j}{r^3} \right). \quad (39)$$

In dyadic form this is

$$\nabla_{\mathbf{k}}^2 \omega^{(0)} = f''(r) \mathbf{e}_r \mathbf{e}_r^\top + \frac{f'(r)}{r} (\mathbf{I} - \mathbf{e}_r \mathbf{e}_r^\top) = f''(r) \mathbf{e}_r \mathbf{e}_r^\top + \frac{f'(r)}{r} \mathbf{e}_\theta \mathbf{e}_\theta^\top. \quad (40)$$

Consequently the orthogonal polar basis diagonalizes the full Cartesian Hessian:

$$(\mathbf{e}_r \quad \mathbf{e}_\theta)^\top \nabla_{\mathbf{k}}^2 \omega^{(0)} (\mathbf{e}_r \quad \mathbf{e}_\theta) = \begin{pmatrix} f''(r) & 0 \\ 0 & f'(r)/r \end{pmatrix}. \quad (41)$$

At $r = k_0$, the ZGV conditions $f'(k_0) = 0$ and $f''(k_0) = a$ reduce Eq. (41) to

$$\nabla_{\mathbf{k}}^2 \omega^{(0)}|_{r=k_0} = a \mathbf{e}_r \mathbf{e}_r^\top, \quad \text{spec} \left(\nabla_{\mathbf{k}}^2 \omega^{(0)}|_{r=k_0} \right) = \{a, 0\}. \quad (42)$$

We now state precisely when the rotationally generated ZGV circle is a Morse–Bott critical manifold. The hypotheses separate the local scalar geometry from the spectral condition that makes a single elastic branch well defined.

Theorem 1.1 (Isotropic ZGV Morse–Bott ring). *Let a lossless homogeneous isotropic plate have a symmetric guided-wave eigenbranch in an open annulus about $r = k_0 > 0$. Assume:*

1. *the eigenvalue is simple throughout that annulus and is separated from every other Lamb and shear-horizontal eigenvalue by a uniform positive mass-normalized eigengap;*
2. *the branch is C^3 in $\mathbf{k}$ and rotational invariance gives $\omega^{(0)}(\mathbf{k}) = f(|\mathbf{k}|)$;*
3. *the regular determinant satisfies $D_S(k_0, \omega_0) = D_{S,k}(k_0, \omega_0) = 0$ and $D_{S,\omega}(k_0, \omega_0) \neq 0$; and*
4. *the radial curvature $a = -D_{S,kk}/D_{S,\omega}$ at (k_0, ω_0) is nonzero.*

Then, in a sufficiently small annular neighbourhood, the critical set of $\omega^{(0)}$ is exactly

$$\Gamma_0 = \{\mathbf{k} \in \mathbb{R}^2 : |\mathbf{k}| = k_0\}. \quad (43)$$

Moreover, at every $\mathbf{k} \in \Gamma_0$,

$$\ker \nabla_{\mathbf{k}}^2 \omega^{(0)} = \text{span}\{\mathbf{e}_\theta\} = T_{\mathbf{k}} \Gamma_0, \quad \nabla_{\mathbf{k}}^2 \omega^{(0)}|_{N_{\mathbf{k}} \Gamma_0} = [a]. \quad (44)$$

Thus Γ_0 is a one-dimensional nondegenerate Morse–Bott critical manifold. If $a > 0$, it is a radial minimum manifold; if $a < 0$, it is a radial maximum manifold.

Proof. The nonzero value of $D_{S,\omega}$ gives a unique local scalar branch by the implicit-function theorem, and the simple uniformly gapped elastic eigenvalue identifies that scalar root with one smooth physical mode. From Eq. (30), $f'(k_0) = 0$, while Eq. (33) gives $f''(k_0) = a \neq 0$. Continuity of f'' permits a sufficiently small $\delta > 0$ such that $f''(r)$ has the sign of a whenever $|r - k_0| < \delta$. Equivalently, the fundamental theorem of calculus gives

$$f'(r) = f'(k_0) + \int_{k_0}^r f''(\xi) d\xi = (r - k_0) \int_0^1 f''(k_0 + \tau(r - k_0)) d\tau. \quad (45)$$

The integral factor is nonzero in that interval, so $f'(r) = 0$ there if and only if $r = k_0$. Equation (38) then proves that the critical set in the annulus is exactly Γ_0 .

For each point on the ring, Eq. (42) maps $\mathbf{e}_r$ to $a\mathbf{e}_r$ and maps $\mathbf{e}_\theta$ to zero. Since $a \neq 0$, its kernel is precisely $\text{span}\{\mathbf{e}_\theta\}$, which is the tangent line to the circle, while the restriction to the radial normal line is the nonsingular one-by-one form $[a]$. These are exactly the critical-set and normal-nondegeneracy conditions in the definition of a Morse–Bott function. The sign of a supplies the final minimum/maximum classification. $\square$

For the reference branch in Eq. (35), $a > 0$, so the normal Morse index is zero. The tangential zero eigenvalue is not an accidental higher-order radial degeneracy: it is enforced by the continuous rotational symmetry. The cases $a = 0$, $D_{S,\omega} = 0$, or a closing modal eigengap violate distinct hypotheses of Theorem 1.1 and require a higher-order or matrix-valued local reduction.

2 Weak-anisotropy normal form and Morse splitting

This section closes the coefficients of the weak-anisotropy normal form directly from the Hermitian elastic eigenproblem. We first derive the eigenvalue sensitivity under arbitrary normalization. We then specialize it to the fixed-density plate and differentiate the mode in a fixed mass gauge. A separate tensor calculation shows why the registered cubic family has only a constant and a fourfold term at first order. Finally, a radial implicit-function reduction and the exact Cartesian Hessian give the number and Morse type of the split critical points.

Throughout, a dagger denotes the Hermitian adjoint and $\lambda = \omega^2$. The in-plane polar basis is

$$\mathbf{n}(\theta) = (\cos \theta, \sin \theta, 0), \quad \mathbf{t}(\theta) = (-\sin \theta, \cos \theta, 0), \quad \mathbf{z} = (0, 0, 1). \quad (46)$$

The radial coordinate is denoted by $k > 0$, and $q = k - k_0$ measures distance from the isotropic ZGV circle.

2.1 General Hermitian-pencil sensitivity

Let a real parameter p enter a Hermitian generalized eigenproblem

$$K(p)\mathbf{u}(p) = \lambda(p)M(p)\mathbf{u}(p), \quad K(p)^\dagger = K(p), \quad M(p)^\dagger = M(p) > 0. \quad (47)$$

Assume that λ is simple and let

$$N := \mathbf{u}^\dagger M \mathbf{u} > 0. \quad (48)$$

No normalization has yet been imposed. Differentiating Eq. (47) gives

$$(K_p - \lambda_p M - \lambda M_p)\mathbf{u} + (K - \lambda M)\mathbf{u}_p = \mathbf{0}. \quad (49)$$

Left multiplication by $\mathbf{u}^\dagger$ removes the last term because $\mathbf{u}^\dagger(K - \lambda M) = \mathbf{0}^\dagger$. Therefore

$$\lambda_p = \frac{\mathbf{u}^\dagger(K_p - \lambda M_p)\mathbf{u}}{N}, \quad \omega_p = \frac{\mathbf{u}^\dagger(K_p - \lambda M_p)\mathbf{u}}{2\omega N}. \quad (50)$$

Both the numerator and denominator acquire the same factor $|c|^2$ under $\mathbf{u} \mapsto c\mathbf{u}$, for any nonzero complex c . Thus Eq. (50) is valid with arbitrary normalization and is also phase invariant. Hermiticity makes its numerator real.

The present perturbation changes stiffness but not density. Moreover, the plate mass form is independent of the in-plane wave vector. Hence the fixed-density specialization is

$$\begin{aligned} M_k &= M_\varepsilon = 0, & \lambda_k &= \frac{\mathbf{u}^\dagger K_k \mathbf{u}}{N}, \\ \lambda_\varepsilon &= \frac{\mathbf{u}^\dagger K_\varepsilon \mathbf{u}}{N}, & V(k, \theta) &:= \omega_\varepsilon = \frac{\lambda_\varepsilon}{2\omega}. \end{aligned} \quad (51)$$

Here every derivative is taken on the tracked simple branch, and K_ε is the stiffness derivative per unit ε .

For completeness, the radial derivative of V can also be written before choosing a normalization. At fixed mass define

$$\begin{aligned} S_\varepsilon &:= \mathbf{u}^\dagger K_\varepsilon \mathbf{u}, & (S_\varepsilon)_k &= \mathbf{u}_k^\dagger K_\varepsilon \mathbf{u} + \mathbf{u}^\dagger K_{k\varepsilon} \mathbf{u} + \mathbf{u}^\dagger K_\varepsilon \mathbf{u}_k, \\ N_k &= \mathbf{u}_k^\dagger M \mathbf{u} + \mathbf{u}^\dagger M \mathbf{u}_k. \end{aligned} \quad (52)$$

Applying the quotient rule to $V = S_\varepsilon/(2\omega N)$, and using $\omega_k = \lambda_k/(2\omega)$, yields

$$\partial_k V = \frac{(S_\varepsilon)_k - S_\varepsilon N_k/N}{2\omega N} - \frac{S_\varepsilon \lambda_k}{4\omega^3 N}. \quad (53)$$

The combination in the first numerator cancels the derivative of any k -dependent scale or phase of $\mathbf{u}$. Equation (53) is therefore normalization invariant, as required by the left-hand side.

2.2 Mass gauge, augmented solve, and the complete radial coefficient

For computation and for a compact reduced-resolvent formula, choose the mass-normalized eigenvector and its parallel-transport phase,

$$\mathbf{u}^\dagger M \mathbf{u} = 1, \quad \mathbf{u}^\dagger M \mathbf{u}_k = 0. \quad (54)$$

The real part of the second condition follows by differentiating the first; the phase of $\mathbf{u}$ is chosen to set its imaginary part to zero. Put $A := K - \lambda M$. Equation (49), specialized to $p = k$, together with the gauge is equivalent to the augmented system

$$\begin{pmatrix} K - \lambda M & M\mathbf{u} \\ \mathbf{u}^\dagger M & 0 \end{pmatrix} \begin{pmatrix} \mathbf{u}_k \\ \alpha \end{pmatrix} = \begin{pmatrix} -(K_k - \lambda_k M)\mathbf{u} \\ 0 \end{pmatrix}. \quad (55)$$

For an exact simple eigenpair the multiplier is $\alpha = 0$. The augmented matrix is nonsingular: a homogeneous solution first gives $\alpha = 0$ after left multiplication of its upper row by $\mathbf{u}^\dagger$; simplicity then gives $\mathbf{x} = c\mathbf{u}$, and the lower row forces $c = 0$.

Let $\{(\lambda_n, \mathbf{u}_n)\}$ be a complete set of generalized eigenpairs with $\mathbf{u}_m^\dagger M \mathbf{u}_n = \delta_{mn}$, and let the selected mode have index zero. Taking the inner product of the differentiated equation with $\mathbf{u}_n$, for $n \neq 0$, gives

$$\begin{aligned} (\lambda_n - \lambda) \mathbf{u}_n^\dagger M \mathbf{u}_k &= -\mathbf{u}_n^\dagger (K_k - \lambda_k M) \mathbf{u} \\ &= -\mathbf{u}_n^\dagger K_k \mathbf{u}. \end{aligned} \quad (56)$$

The gauge removes the component along $\mathbf{u}$, so

$$\mathbf{u}_k = \sum_{n \neq 0} \mathbf{u}_n \frac{\mathbf{u}_n^\dagger K_k \mathbf{u}}{\lambda - \lambda_n}. \quad (57)$$

Equivalently, define the reduced resolvent

$$\mathcal{R}_\lambda := \sum_{n \neq 0} \frac{\mathbf{u}_n \mathbf{u}_n^\dagger}{\lambda_n - \lambda}. \quad (58)$$

Then $\mathbf{u}_k = -\mathcal{R}_\lambda (K_k - \lambda_k M) \mathbf{u}$. The modal sum and the augmented system are two representations of the same gauge-fixed derivative. On a compact parameter set, a uniform positive eigengap together with uniform coercivity and boundedness of the mass matrix bounds $\mathcal{R}_\lambda$ in the associated operator norms. Here the mass matrix is fixed and positive definite. Loss of either hypothesis is addressed below as a theorem boundary.

In the gauge (54), differentiation of $\lambda_\varepsilon = \mathbf{u}^\dagger K_\varepsilon \mathbf{u}$ gives the full mixed eigenvalue derivative

$$\lambda_{k\varepsilon} = \mathbf{u}^\dagger K_{k\varepsilon} \mathbf{u} + 2 \operatorname{Re}(\mathbf{u}^\dagger K_\varepsilon \mathbf{u}_k). \quad (59)$$

Consequently the radial coefficient in the frequency normal form is

$$\begin{aligned} B(\theta) &:= \partial_k \partial_\varepsilon \omega|_{(k_0, \theta, 0)} = \left(\frac{\lambda_{k\varepsilon}}{2\omega} - \frac{\lambda_\varepsilon \lambda_k}{4\omega^3} \right) \Big|_{(k_0, \theta, 0)} \\ &= \left[\frac{\mathbf{u}^\dagger K_{k\varepsilon} \mathbf{u}}{2\omega} + \frac{\operatorname{Re}(\mathbf{u}^\dagger K_\varepsilon \mathbf{u}_k)}{\omega} - \frac{\lambda_\varepsilon \lambda_k}{4\omega^3} \right] \Big|_{(k_0, \theta, 0)}. \end{aligned} \quad (60)$$

The three displayed contributions are, respectively, the explicit mixed stiffness derivative, the differentiated-mode contribution, and the frequency-denominator contribution. At the exact isotropic ZGV point, $\lambda_k = 2\omega\omega_k = 0$, so the last contribution evaluates to zero there. It remains part of the general identity and is required away from the exact ZGV radius. In particular, B is not a fixed-mode strain integral.

2.3 Rotation and complete contraction of the cubic perturbation

Let $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$ be the orthonormal cubic axes. A cubic elasticity tensor can be decomposed exactly as

$$\begin{aligned} C_{ijkl} &= C_{12} \delta_{ij} \delta_{kl} + C_{44} (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}) \\ &\quad + \Delta_C \sum_{\alpha=1}^3 a_{\alpha i} a_{\alpha j} a_{\alpha k} a_{\alpha l}, \quad \Delta_C = C_{11} - C_{12} - 2C_{44}. \end{aligned} \quad (61)$$

Indeed, the $iiii$, $iiij$, and $ijij$ components of the right-hand side are $C_{12} + 2C_{44} + \Delta_C = C_{11}$, C_{12} , and C_{44} , respectively, and cubic symmetry generates all remaining nonzero components.

The registered bulk-preserving family is

$$C_{11} = \lambda + 2\mu + \frac{2}{3}\varepsilon\delta, \quad C_{12} = \lambda - \frac{1}{3}\varepsilon\delta, \quad C_{44} = \mu. \quad (62)$$

It has

$$C_{12}^{(1)} = -\frac{\delta}{3}, \quad C_{44}^{(1)} = 0, \quad \Delta_C^{(1)} := \partial_\varepsilon \Delta_C|_0 = \delta, \quad \Delta_C(\varepsilon) = \varepsilon\delta. \quad (63)$$

Thus Eq. (61) gives

$$C_{ijkl}^{(1)} = -\frac{\delta}{3}\delta_{ij}\delta_{kl} + \delta \sum_{\alpha=1}^3 a_{\alpha i} a_{\alpha j} a_{\alpha k} a_{\alpha l}. \quad (64)$$

The cubic stability inequalities reduce to $\mu > 0$, $2\mu + \varepsilon\delta > 0$, and $3\lambda + 2\mu > 0$. Thus, for the registered $\delta > 0$, stability requires $\varepsilon > -2\mu/\delta$; the weak signed interval used below is chosen as a compact neighbourhood of zero inside this bound. No assertion is made for arbitrarily large negative ε .

For a [001]-cut plate, take the crystallographic axes to be $\mathbf{a}_1 = \mathbf{e}_x$, $\mathbf{a}_2 = \mathbf{e}_y$, and $\mathbf{a}_3 = \mathbf{e}_z$. The rotation whose columns are $(\mathbf{n}, \mathbf{t}, \mathbf{z})$ is

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \tilde{C}_{abcd}^{(1)}(\theta) = R_{ia} R_{jb} R_{kc} R_{ld} C_{ijkl}^{(1)}. \quad (65)$$

The isotropic sagittal mode propagating along $\mathbf{n}$ has displacement

$$\hat{\mathbf{u}}(z; \theta) = U(z)\mathbf{n} + W(z)\mathbf{z}, \quad (66)$$

and strain

$$\begin{aligned} \boldsymbol{\epsilon} &= E \mathbf{n}\mathbf{n} + Z \mathbf{z}\mathbf{z} + S(\mathbf{n}\mathbf{z} + \mathbf{z}\mathbf{n}), \\ E &= ik_0 U, \quad Z = W', \quad S = \frac{U' + ik_0 W}{2}. \end{aligned} \quad (67)$$

Writing $c = \cos \theta$ and $s = \sin \theta$, its six independent components in the crystal frame are

$$\epsilon_{xx} = c^2 E, \quad \epsilon_{yy} = s^2 E, \quad \epsilon_{zz} = Z, \quad \epsilon_{xy} = csE, \quad \epsilon_{xz} = cS, \quad \epsilon_{yz} = sS. \quad (68)$$

Before inserting the special family, the complete cubic strain contraction is

$$\begin{aligned} \epsilon_{ij}^* C_{ijkl}^{(1)} \epsilon_{kl} &= C_{11}^{(1)} (|\epsilon_{xx}|^2 + |\epsilon_{yy}|^2 + |\epsilon_{zz}|^2) \\ &\quad + 2C_{12}^{(1)} \operatorname{Re}(\epsilon_{xx}^* \epsilon_{yy} + \epsilon_{xx}^* \epsilon_{zz} + \epsilon_{yy}^* \epsilon_{zz}) \\ &\quad + 4C_{44}^{(1)} (|\epsilon_{xy}|^2 + |\epsilon_{xz}|^2 + |\epsilon_{yz}|^2). \end{aligned} \quad (69)$$

Every normal and shear component in Eq. (68) is represented here. The required trigonometric products are

$$c^4 + s^4 = \frac{3 + \cos 4\theta}{4}, \quad c^2 s^2 = \frac{1 - \cos 4\theta}{8}. \quad (70)$$

Substituting Eqs. (63), (68), and (70) into Eq. (69) gives, term by term,

$$\epsilon^* : C^{(1)} : \epsilon = \delta \left[\frac{5}{12} |E|^2 + \frac{2}{3} |Z|^2 - \frac{2}{3} \operatorname{Re}(E^* Z) + \frac{1}{4} |E|^2 \cos 4\theta \right]. \quad (71)$$

The shear strain S has zero coefficient here precisely because $C_{44}^{(1)} = 0$, not because it was excluded from the contraction.

For a mode of arbitrary mass norm

$$N = \int_{-h}^h \rho(|U|^2 + |W|^2) dz, \quad (72)$$

the first-order sensitivity is therefore

$$V(\theta) = V_0 + V_4 \cos 4\theta, \quad (73)$$

where

$$V_0 = \frac{\delta}{2\omega_0 N} \int_{-h}^h \left[\frac{5}{12} |E|^2 + \frac{2}{3} |Z|^2 - \frac{2}{3} \operatorname{Re}(E^* Z) \right] dz, \quad (74)$$

$$V_4 = Q_4 \Delta_C^{(1)}, \quad Q_4 := \frac{1}{8\omega_0 N} \int_{-h}^h |E|^2 dz = \frac{k_0^2}{8\omega_0 N} \int_{-h}^h |U|^2 dz > 0. \quad (75)$$

The last inequality holds for the selected finite-wavenumber Lamb mode, whose in-plane component is nonzero. Since $\Delta_C^{(1)} = \delta > 0$ in the current family, it follows analytically that $V_4 > 0$. The reference spectral evaluation gives $V_4 = 0.014063828174045103 > 0$ in the registered dimensionless units. The physical angular frequency displacement is

$$\varepsilon V_4 = Q_4 \Delta_C(\varepsilon), \quad (76)$$

so ε is applied exactly once. Cubic symmetry by itself would allow $\cos(8\theta)$ and further multiples of four; the explicit first-order tensor contraction proves that they are absent for this linear perturbation family.

2.4 Controlled normal form and radial implicit-function reduction

We now state the local reduction with its derivative-level remainder. The regularity assumptions are stronger than the minimum needed for the radial root because later stationary-phase arguments differentiate the phase and amplitude further.

Theorem 2.1 (Weak-anisotropy normal form). *Let the isotropic plate possess a positive-frequency ZGV circle at $k = k_0 > 0$, with radial curvature $a \neq 0$. Assume that, in a closed annulus about this circle and for $|\varepsilon| \leq \varepsilon_0$, the tracked generalized eigenvalue is simple, the mass matrix is positive definite, and the mass-normalized eigengap obeys*

$$\inf_{\theta, q, \varepsilon} \min_{n \neq 0} |\lambda_n - \lambda| = g > 0. \quad (77)$$

Assume also that the Hermitian pencil is sufficiently smooth that the selected real frequency branch is C^6 in the local variables. The elastic pencil used here is analytic, so this hypothesis is automatic away from an eigengap closure. Then, after reducing the annulus if necessary,

$$\omega(q, \theta; \varepsilon) = \omega_0 + \frac{a}{2} q^2 + \varepsilon [V(\theta) + qB(\theta)] + R(q, \theta, \varepsilon), \quad (78)$$

where $V = \partial_\varepsilon \omega(k_0, \theta; 0)$, $B = \partial_k \partial_\varepsilon \omega(k_0, \theta; 0)$, and

$$R = q^3 r_{30} + \varepsilon q^2 r_{12} + \varepsilon^2 r_{02}, \quad (79)$$

with uniformly bounded periodic C^3 coefficient functions. In particular, with $(x)_+ = \max(x, 0)$, every derivative used in the two-dimensional Morse reduction satisfies

$$\left| \partial_q^i \partial_\theta^j R(q, \theta, \varepsilon) \right| \leq C_{ij} \left(|q|^{3-i} + |\varepsilon| |q|^{(2-i)_+} + \varepsilon^2 \right), \quad 0 \leq i + j \leq 2. \quad (80)$$

The same bound with $i = 1$ holds after two angular derivatives, as follows directly from Eq. (79). There is a unique periodic radial stationary graph $q = q_*(\theta, \varepsilon)$ near zero, and

$$q_*(\theta, \varepsilon) = -\frac{\varepsilon B(\theta)}{a} + O_{C^2}(\varepsilon^2). \quad (81)$$

If $W_2(\theta) := r_{02}(0, \theta, 0) = \frac{1}{2} \partial_\varepsilon^2 \omega(k_0, \theta; 0)$, the frequency restricted to this graph is

$$\omega_*(\theta; \varepsilon) := \omega(q_*(\theta, \varepsilon), \theta; \varepsilon) = \omega_0 + \varepsilon V(\theta) + \varepsilon^2 \left[W_2(\theta) - \frac{B(\theta)^2}{2a} \right] + O_{C^2}(\varepsilon^3). \quad (82)$$

Proof. The uniform eigengap and simplicity make the spectral projector and the selected eigenvalue smooth. Taylor expansion first in q and then in ε , using $\omega_q(0, \theta; 0) = 0$, $\omega_{qq}(0, \theta; 0) = a$, and isotropic independence of θ , gives Eqs. (78) and (79). Applying the product rule to the three factored terms gives Eq. (80) derivative by derivative.

Radial stationarity is the equation

$$F(q, \theta, \varepsilon) := \omega_q = aq + \varepsilon B(\theta) + R_q = 0. \quad (83)$$

At $(q, \varepsilon) = (0, 0)$, one has $F = 0$ and $F_q = a \neq 0$. The parameter-dependent implicit-function theorem, applied uniformly on the compact angular circle, gives the unique periodic graph. It first gives $q_* = O(\varepsilon)$. The $i = 1$ bound in Eq. (80) then gives $R_q(q_*, \theta, \varepsilon) = O_{C^2}(\varepsilon^2)$. Solving Eq. (83) for q_* proves Eq. (81).

At $q_* = O(\varepsilon)$, the three parts of the remainder are

$$q_*^3 r_{30} = O_{C^2}(\varepsilon^3), \quad \varepsilon q_*^2 r_{12} = O_{C^2}(\varepsilon^3), \quad \varepsilon^2 r_{02} = \varepsilon^2 W_2 + O_{C^2}(\varepsilon^3). \quad (84)$$

Finally,

$$\frac{a}{2} q_*^2 + \varepsilon q_* B = -\frac{\varepsilon^2 B^2}{2a} + O_{C^2}(\varepsilon^3), \quad (85)$$

which proves Eq. (82). $\square$

2.5 Exact Cartesian Hessian and its Schur inertia

The Morse classification must use the Cartesian Hessian, not the matrix of uncorrected polar second partial derivatives. For any scalar $f(q, \theta)$, with $k = k_0 + q > 0$, direct differentiation of $\nabla_{\mathbf{k}} f = \mathbf{n}f_q + \mathbf{t}f_\theta/k$ gives the Cartesian Hessian in the orthonormal polar basis:

$$(\mathbf{n} \ \mathbf{t})^\top \nabla_{\mathbf{k}}^2 f (\mathbf{n} \ \mathbf{t}) = \begin{pmatrix} f_{qq} & k^{-1}f_{q\theta} - k^{-2}f_\theta \\ k^{-1}f_{q\theta} - k^{-2}f_\theta & k^{-1}f_q + k^{-2}f_{\theta\theta} \end{pmatrix}. \quad (86)$$

At a critical point (q_j, θ_j) , both first partial derivatives vanish, and the exact matrix is

$$\mathbf{H}_j = \begin{pmatrix} \omega_{qq} & k_j^{-1}\omega_{q\theta} \\ k_j^{-1}\omega_{q\theta} & k_j^{-2}\omega_{\theta\theta} \end{pmatrix}_{(q_j, \theta_j)}, \quad k_j = k_0 + q_j. \quad (87)$$

Along the radial graph, implicit differentiation gives

$$q'_* = -\frac{\omega_{q\theta}}{\omega_{qq}}, \quad \omega'_* = \omega_\theta, \quad \omega''_* = \omega_{\theta\theta} - \frac{\omega_{q\theta}^2}{\omega_{qq}}. \quad (88)$$

If $\omega_{qq} \neq 0$, Sylvester's law of inertia applied to Eq. (87) gives the exact decomposition

$$\text{inertia}(\mathbf{H}_j) = \text{inertia}(\omega_{qq}) + \text{inertia}\left(\frac{\omega''_*}{k_j^2}\right). \quad (89)$$

Thus the mixed derivative changes the tangential eigenvalue through the Schur complement; it cannot be discarded before the inertia is taken.

At a critical point generated by the normal form, Eqs. (80) and (81) give

$$\mathbf{H}_j = \begin{pmatrix} a + O(\varepsilon) & \frac{\varepsilon B'(\theta_j)}{k_0} + O(\varepsilon^2) \\ \frac{\varepsilon B'(\theta_j)}{k_0} + O(\varepsilon^2) & \frac{\varepsilon V''(\theta_j)}{k_0^2} + O(\varepsilon^2) \end{pmatrix}. \quad (90)$$

Its Schur complement is

$$\frac{1}{k_j^2} \left(\omega_{\theta\theta} - \frac{\omega_{q\theta}^2}{\omega_{qq}} \right) = \frac{\varepsilon V''(\theta_j)}{k_0^2} + O(\varepsilon^2). \quad (91)$$

For the reference branch $a > 0$, the radial inertia is positive and the sign of $\varepsilon V''$ determines whether the second direction is positive or negative.

2.6 General even harmonic and the cubic eight-point theorem

Reciprocity gives $\omega(k, \theta; \varepsilon) = \omega(k, \theta + \pi; \varepsilon)$. Therefore the order m of the first nonconstant angular harmonic must be even. A mirror axis fixes the phase origin of each allowed harmonic, but does not by itself remove other harmonics. We now impose the additional pure-leading-harmonic hypothesis

$$V(\theta) = V_0 + V_m \cos[m(\theta - \theta_0)], \quad V_m \neq 0. \quad (92)$$

Its derivative has the $2m$ simple zeros

$$\theta_j^{(0)} = \theta_0 + \frac{j\pi}{m}, \quad j = 0, \dots, 2m-1, \quad (93)$$

and

$$V''(\theta_j^{(0)}) = -m^2 V_m (-1)^j. \quad (94)$$

Since $\omega'_* = \varepsilon V' + O_{C^1}(\varepsilon^2)$, the ordinary implicit-function theorem at each simple zero gives

$$\theta_j = \theta_0 + \frac{j\pi}{m} + O(\varepsilon), \quad k_j = k_0 - \frac{\varepsilon B(\theta_j)}{a} + O(\varepsilon^2). \quad (95)$$

For $a > 0$, Eq. (91) shows that the points alternate between m minima and m saddles. Reversing the sign of εV_m leaves the leading angular locations fixed and exchanges their roles. Their leading frequencies are

$$\omega_j = \omega_0 + \varepsilon[V_0 + V_m(-1)^j] + O(\varepsilon^2), \quad \Delta\omega_{\text{ms}} = 2|\varepsilon V_m| + O(\varepsilon^2). \quad (96)$$

The gradient index is $+1$ at each minimum and -1 at each saddle, so the algebraic index sum is

$$m(+1) + m(-1) = 0. \quad (97)$$

A fixed higher harmonic must be distinguished from the $O(\varepsilon)$ correction to the roots above. Suppose instead that

$$V(\theta) = V_0 + V_m \cos[m(\theta - \theta_0)] + \mathcal{H}(\theta). \quad (98)$$

There is a constant $c_m > 0$ such that the dominance condition

$$\|\mathcal{H}\|_{C^2} < c_m |V_m| \quad (99)$$

preserves the $2m$ simple zeros and their alternating curvatures. Their locations obey the separate estimate

$$\theta_j = \theta_0 + \frac{j\pi}{m} + O\left(\frac{\|\mathcal{H}\|_{C^1}}{|V_m|}\right) + O(\varepsilon). \quad (100)$$

Without a bound such as Eq. (99), point-group symmetry alone does not fix the number of angular critical points.

Theorem 2.2 (Cubic Morse splitting). *Assume the hypotheses of Theorem 2.1, take the reference radial curvature $a > 0$, and use the [001] cubic family (62) with $\delta > 0$. Then, for every sufficiently small nonzero ε , a thin annular neighbourhood of the isotropic ZGV circle contains exactly eight critical points of the tracked branch. They satisfy*

$$\theta_j = \frac{j\pi}{4} + O(\varepsilon), \quad k_j = k_0 - \frac{\varepsilon B(\theta_j)}{a} + O(\varepsilon^2), \quad j = 0, \dots, 7. \quad (101)$$

Four are Morse minima and four are Morse saddles. For $\varepsilon > 0$, the odd j are minima and the even j are saddles; for $\varepsilon < 0$, these roles reverse. The two critical-frequency families are separated by

$$\Delta\omega_{\text{ms}} = 2|\varepsilon V_4| + O(\varepsilon^2), \quad V_4 = Q_4 \delta > 0, \quad (102)$$

and their total gradient index is zero.

Proof. Equations (71)–(75) prove, rather than infer from symmetry, that the first-order angular sensitivity is the pure harmonic $V_0 + V_4 \cos 4\theta$ and that $V_4 > 0$. Set $m = 4$ and $\theta_0 = 0$ in Eqs. (93)–(95). This gives eight points. At their leading locations,

$$\varepsilon V''(\theta_j) = -16\varepsilon V_4(-1)^j + O(\varepsilon^2). \quad (103)$$

The radial Schur pivot is positive, while the second pivot has the sign in Eq. (103). The stated minimum–saddle classification and its reversal follow. Equation (96) supplies the separation, and Eq. (97) supplies the index sum. To prove that the count is exact, choose disjoint angular neighbourhoods of the eight simple zeros of V' . On their compact complement, $|V'|$ has a positive lower bound. After division by the nonzero ε , the $O_{C^1}(\varepsilon)$ correction in ω'_*/ε cannot create a zero there when $|\varepsilon|$ is sufficiently small. The radial graph is unique at every angle by Theorem 2.1; hence no additional critical point exists in the reduced annulus. $\square$

2.7 Exceptional cases and theorem boundaries

The condition $V_m \neq 0$ is substantive. If a nominal coefficient such as V_4 vanishes, the following cases must be distinguished.

1. If a higher harmonic $V_\ell \cos(\ell\theta)$, with even $\ell > m$, is the first nonzero nonconstant term at order ε , the preceding calculation applies with m replaced by ℓ .
2. If the whole first-order sensitivity is angularly constant, the second order reduced coefficient is

$$V_2^{\text{eff}}(\theta) := W_2(\theta) - \frac{B(\theta)^2}{2a}. \quad (104)$$

A nonzero nondegenerate harmonic of V_2^{eff} splits the ring with tangential curvature of order ε^2 , not order ε .

3. If two or more harmonics compete at the same first nonzero order, the critical-point count is the zero count of their full derivative; it is not determined by one nominal order. If at a root both $V' = 0$ and $V'' = 0$, the point is degenerate and the Morse conclusion does not apply until the next local term is resolved.

Two independent spectral boundaries are equally important. If $a = 0$, then $F_q(0, \theta, 0) = 0$, the radial implicit-function argument fails, and the balance between anisotropy and the first nonzero higher radial term has a different scaling. If the eigengap g closes, the reduced resolvent becomes unbounded and a scalar tracked branch is not smooth; a matrix-valued degenerate perturbation problem is then required. The cases $k_0 = 0$ and $\omega_0 = 0$ are also outside the present polar-ring and positive-square-root formulation, respectively.

2.8 Conversion to the dimensionless computational convention

The code uses

$$\kappa = kh, \quad \Omega = \frac{\omega h}{c_T}, \quad c_T = \sqrt{\frac{\mu}{\rho}}, \quad q_\kappa = \kappa - \kappa_0 = hq. \quad (105)$$

Writing subscripts ω and Ω for coefficients in the dimensional and dimensionless frequency normal forms, the chain rule gives

$$V_\Omega = \frac{h}{c_T} V_\omega, \quad B_\Omega = \frac{1}{c_T} B_\omega, \quad a_\Omega = \frac{1}{hc_T} a_\omega. \quad (106)$$

Hence

$$q_{\kappa,*} = -\frac{\varepsilon B_\Omega}{a_\Omega} + O(\varepsilon^2) = h \left[-\frac{\varepsilon B_\omega}{a_\omega} + O(\varepsilon^2) \right], \quad (107)$$

which is exactly the dimensional location multiplied by h . If the dimensional invariant $\Delta_C^{(1)}$ is retained, then $Q_{4,\Omega} = (h/c_T)Q_{4,\omega}$. If stiffness is additionally divided by μ , the corresponding coefficient absorbs the factor μ . For the registered reference values $h = \rho = \mu = 1$, one has $c_T = 1$, so the numerical and dimensional coefficient values coincide in the chosen units.

3 Bessel–Morse asymptotics of the Green response

We derive the time-domain response of the ZGV branch with every Fourier, modal, and stationary-phase constant retained. The object propagated below is the positive-frequency complex response. It is not doubled. The physical real response is reconstructed only after the branch integral has been evaluated. This distinction fixes a factor of two that is otherwise easy to hide in a loosely defined analytic signal.

Throughout this section $a > 0$, $k_0 > 0$, and the time dependence of a positive-frequency mode is $\exp(-i\omega t)$. The hypotheses on branch simplicity, eigengap, and smoothness are those of Theorem 2.1. All asymptotic statements concern the selected positive-frequency branch in a fixed annular spectral window; contributions from other branches are separate terms in the complete Green function.

3.1 Fourier convention and normalization-invariant branch projection

For an in-plane field, use the transform pair

$$\hat{\mathbf{u}}(\mathbf{k}, z, t) = \int_{\mathbb{R}^2} \mathbf{u}(\mathbf{x}, z, t) e^{-i\mathbf{k} \cdot \mathbf{x}} d^2\mathbf{x}, \quad \mathbf{u}(\mathbf{x}, z, t) = \mathcal{F}_2 \int_{\mathbb{R}^2} \hat{\mathbf{u}}(\mathbf{k}, z, t) e^{i\mathbf{k} \cdot \mathbf{x}} d^2\mathbf{k}, \quad \mathcal{F}_2 = (2\pi)^{-2}. \quad (108)$$

At fixed $\mathbf{k}$, let the thickness-discretized or weak-form dynamics under an impulsive generalized load be

$$M\ddot{\mathbf{U}} + K\mathbf{U} = \mathbf{f} \delta(t), \quad K\mathbf{u} = \omega^2 M\mathbf{u}, \quad N := \mathbf{u}^\dagger M\mathbf{u} > 0. \quad (109)$$

The vector $\mathbf{f} = \mathbf{f}(\mathbf{k})$ includes the through-thickness source functional, and a linear detector is written $D[\mathbf{U}] = \mathbf{d}^\dagger \mathbf{U}$. Projecting Eq. (109) onto a simple mode gives

$$\ddot{q} + \omega^2 q = \frac{\mathbf{u}^\dagger \mathbf{f}}{N} \delta(t), \quad q(t) = H(t) \frac{\mathbf{u}^\dagger \mathbf{f}}{\omega N} \sin(\omega t). \quad (110)$$

Because

$$\frac{\sin(\omega t)}{\omega} = 2 \operatorname{Re} \left[\frac{i}{2\omega} e^{-i\omega t} \right], \quad (111)$$

the undoubled coefficient of the positive-frequency exponential in the detected response is

$$\mathcal{P}(\mathbf{k}) := \frac{i}{2\omega} \frac{(\mathbf{d}^\dagger \mathbf{u})(\mathbf{u}^\dagger \mathbf{f})}{\mathbf{u}^\dagger M \mathbf{u}}. \quad (112)$$

This source–mode–detector amplitude is invariant under an arbitrary nonzero complex normalization $\mathbf{u} \mapsto c(\mathbf{k})\mathbf{u}$: the numerator and denominator both acquire $|c|^2$, including when the scale and phase depend on $\mathbf{k}$. Thus no mass-normalization convention enters the observable amplitude.

In the mass-normalized, colocated top-face normal channel, $\mathbf{d} = \mathbf{f} = \mathbf{e}_z(h)$ and $\mathbf{u}^\dagger M \mathbf{u} = 1$. The general projection then reduces to the phase-invariant overlap used by the numerical implementation,

$$\mathcal{P}_n(\mathbf{k}) = \frac{i}{2\omega_n} |u_{n,z}(h; \mathbf{k})|^2. \quad (113)$$

Let $G^+(t)$ denote the inverse Fourier integral made from the coefficient in Eq. (112). For a real load and a real detector, the negative-frequency term is its complex conjugate after the reciprocal wave-vector partner is included. Consequently

$$G_{\text{phys}}(t) = G^+(t) + G^+(t)^* = 2 \operatorname{Re} G^+(t). \quad (114)$$

Equation (114), not an extra factor inside G^+ , defines the physical real response. Every theorem below is first stated for G^+ , and the physical result follows from this equation.

For reference, a normalized circular Gaussian source of radius r_s ,

$$s(\mathbf{x}) = \frac{1}{\pi r_s^2} \exp\left(-\frac{|\mathbf{x}|^2}{r_s^2}\right), \quad \hat{s}(k) = \exp\left[-\left(\frac{r_s k}{2}\right)^2\right], \quad (115)$$

multiplies $\mathcal{P}$ by $\hat{s}(k)$. The numerical local-branch window is the smooth super-Gaussian multiplied by the compact taper used by the direct numerical integral,

$$\chi_{\sigma_q}(q) = \exp[-(q/\sigma_q)^8] b_{\sigma_q}(q), \quad q = k - k_0, \quad (116)$$

although an ordinary Gaussian window $\exp[-q^2/(2\sigma_q^2)]$ gives the same leading asymptotics. Only smoothness, localization inside the annulus, and a nonzero value at the stationary set are used in the proof. Here $b_{\sigma_q} = 1$ on $|q| \leq 1.25\sigma_q$, $b_{\sigma_q} = 0$ on $|q| \geq 1.5\sigma_q$, and the intervening strip uses the standard flat step constructed from $\exp(-1/x)$. Hence b_{σ_q} is C^∞ , all endpoint derivatives vanish, every local coefficient below is unchanged, and the numerical integral satisfies the same no-endpoint hypothesis as the theorem.

Define the complete projected amplitude, excluding the Fourier measure and the polar Jacobian, by

$$\mathcal{A}(q, \theta; \varepsilon) := \chi_{\sigma_q}(q) \exp\left[-\left(\frac{r_s(k_0 + q)}{2}\right)^2\right] \frac{i}{2\omega} \frac{(\mathbf{d}^\dagger \mathbf{u})(\mathbf{u}^\dagger \mathbf{f}_z)}{\mathbf{u}^\dagger M \mathbf{u}}. \quad (117)$$

Here $\mathbf{f}_z$ is the unit through-thickness source functional; any additional calibrated source strength can be included multiplicatively. For the colocated axisymmetric normal channel in the isotropic plate,

$$\mathcal{A}(0, \theta; 0) = A_0, \quad A_0 \neq 0. \quad (118)$$

In particular, A_0 does not contain k_0 , $\mathcal{F}_2$, or the angular factor 2π .

3.2 Local polar integral and radial stationary phase

Let $0 < \eta < k_0$ be small enough that the selected branch remains simple on $|q| \leq \eta$, and let the window vanish smoothly before the annular boundary if a compact cutoff is needed. The exact local positive-frequency complex response is

$$G^+(t; \varepsilon) = \mathcal{F}_2 \int_0^{2\pi} \int_{-\eta}^{\eta} \mathcal{A}(q, \theta; \varepsilon) \exp[-it\omega(q, \theta; \varepsilon)] (k_0 + q) dq d\theta. \quad (119)$$

This displays the complete Fourier/polar measure $(2\pi)^{-2}(k_0 + q) dq d\theta$.

For sufficiently small $|\varepsilon|$, Theorem 2.1 supplies the unique radial stationary graph $q = q_*(\theta, \varepsilon)$. Uniform one-dimensional stationary phase in q , with the angular variable treated as a compact parameter, gives

$$G^+(t; \varepsilon) = \mathcal{F}_2 e^{-i\pi/4} \left(\frac{2\pi}{t} \right)^{1/2} \int_0^{2\pi} \frac{(k_0 + q_*) \mathcal{A}(q_*, \theta; \varepsilon)}{\sqrt{\omega_{qq}(q_*, \theta; \varepsilon)}} e^{-it\omega_*(\theta; \varepsilon)} d\theta + O(t^{-3/2}), \quad (120)$$

$$\omega_*(\theta; \varepsilon) = \omega_0 + \varepsilon V(\theta) + \varepsilon^2 W_2^{\text{eff}}(\theta) + O_{C^2}(\varepsilon^3),$$

$$W_2^{\text{eff}}(\theta) := W_2(\theta) - \frac{B(\theta)^2}{2a}. \quad (121)$$

The last line is exactly the radial-relaxation correction derived in Eq. (82). Keeping W_2 but dropping $-B^2/(2a)$ would give an incorrect second-order phase and, therefore, an incorrect overlap window.

The radial onset time follows by requiring the stationary width $(at)^{-1/2}$ to be small compared with the window width σ_q :

$$t_{\text{rad}} \asymp \frac{1}{a\sigma_q^2}. \quad (122)$$

The Fresnel phase in Eq. (120) follows from the exact continuation

$$\int_{-\infty}^{\infty} \exp\left(-\frac{i}{2}atq^2\right) dq = e^{-i\pi/4} \sqrt{\frac{2\pi}{at}}, \quad a > 0, \quad t > 0, \quad (123)$$

where the oscillatory integral is defined by Gaussian continuation:

$$\lim_{\eta \downarrow 0} \int_{-\infty}^{\infty} \exp\left[-\left(\eta + \frac{i}{2}at\right)q^2\right] dq = e^{-i\pi/4} \sqrt{\frac{2\pi}{at}}. \quad (124)$$

3.3 The joint transition limit and its exact Bessel kernel

For the [001] cubic family, $V(\theta) = V_0 + V_4 \cos 4\theta$, with V_4 the sensitivity coefficient and εV_4 the physical frequency shift. Introduce

$$\tau := \varepsilon V_4 t. \quad (125)$$

The sign of τ is retained; the resulting J_0 kernel is even.

Theorem 3.1 (Uniform Bessel crossover of the ZGV ring). *Assume that the exact phase is C^6 , the amplitude is C^4 , the radial curvature satisfies $a > 0$, $V_4 \neq 0$, and Eq. (118) holds. In the joint limit*

$$t \longrightarrow \infty, \quad \varepsilon \longrightarrow 0, \quad \tau = \varepsilon V_4 t \in K, \quad (126)$$

where $K \subset \mathbb{R}$ is any fixed compact set,

$$G^+(t; \varepsilon) = \mathcal{C} e^{-i(\omega_0 + \varepsilon V_0)t} t^{-1/2} J_0(\varepsilon V_4 t) + o(t^{-1/2}), \quad (127)$$

uniformly for $\tau \in K$, with the fully determined constant

$$\mathcal{C} = \mathcal{F}_2(2\pi) k_0 A_0 e^{-i\pi/4} \sqrt{\frac{2\pi}{a}}. \quad (128)$$

No amplitude, phase, carrier-frequency correction, or time shift is fitted in Eqs. (127)–(128). The corresponding physical real response is $G_{\text{phys}} = 2 \operatorname{Re} G^+$.

Proof. Under Eq. (126), one has $\varepsilon = O(t^{-1})$. Equations (81) and (121) therefore imply, uniformly in θ ,

$$\begin{aligned} q_* &= O(t^{-1}), & t\varepsilon^2 W_2^{\text{eff}} &= O(t^{-1}), & tO(\varepsilon^3) &= O(t^{-2}), \\ \frac{(k_0 + q_*) \mathcal{A}(q_*, \theta; \varepsilon)}{\sqrt{\omega_{qq}(q_*, \theta; \varepsilon)}} &= \frac{k_0 A_0}{\sqrt{a}} + O(t^{-1}). \end{aligned} \quad (129)$$

Substitution into Eq. (120) leaves the angular identity

$$\int_0^{2\pi} e^{-i\tau \cos 4\theta} d\theta = 4 \int_0^{\pi/2} e^{-i\tau \cos 4\theta} d\theta = \int_0^{2\pi} e^{-i\tau \cos \phi} d\phi = 2\pi J_0(\tau). \quad (130)$$

Here the change $\phi = 4\theta$ covers four identical periods. Combining Eq. (130) with the radial Fresnel factor gives Eq. (128) exactly. The error is uniform in absolute size on compact τ -sets. At a zero of J_0 , the displayed leading term vanishes, so the theorem does not assert a uniform relative error after division by J_0 . $\square$

3.4 General angular weights and the Fourier–Bessel selection rule

Axisymmetry is not required for a uniform angular representation. Let the leading amplitude on the ring have the absolutely convergent Fourier series

$$A(\theta) = \sum_{\ell \in \mathbb{Z}} A_\ell e^{i\ell\theta}, \quad A_\ell = \frac{1}{2\pi} \int_0^{2\pi} A(\theta) e^{-i\ell\theta} d\theta. \quad (131)$$

The Jacobi–Anger expansion, in the sign convention used here, is

$$e^{-i\tau \cos 4\theta} = \sum_{n \in \mathbb{Z}} (-i)^n J_n(\tau) e^{i4n\theta}. \quad (132)$$

Define the normalized angular kernel

$$K_A(\tau) := \frac{1}{2\pi} \int_0^{2\pi} A(\theta) e^{-i\tau \cos 4\theta} d\theta. \quad (133)$$

Termwise integration yields the exact selection rule

$$K_A(\tau) = \sum_{n \in \mathbb{Z}} A_{-4n} (-i)^n J_n(\tau) = \sum_{\substack{\ell \in \mathbb{Z} \\ 4|\ell}} A_\ell (-i)^{-\ell/4} J_{-\ell/4}(\tau). \quad (134)$$

Thus only amplitude harmonics divisible by four survive the full-period integral at this order. The canonical law is the $A_0 J_0$ term. If $A_0 = 0$ but another selected coefficient is nonzero, the leading uniform kernel is the fixed linear combination of compatible higher-order Bessel functions prescribed by Eq. (134). It reduces to one higher-order Bessel function only when a single compatible harmonic pair is present; it is not legitimate to restore a J_0 term by fitting a complex constant. The numerical weighted-kernel routine evaluates precisely K_A . The factor 2π from the unnormalized angular integral already appears once, and only once, in $\mathcal{C}$ in Eq. (128).

For a pure even harmonic of general order m , the same change of variable gives

$$\int_0^{2\pi} e^{-i\tau \cos[m(\theta - \theta_0)]} d\theta = 2\pi J_0(\tau), \quad m \in 2\mathbb{N}. \quad (135)$$

The independence of the constant from m will reappear as an exact cancellation between Morse-point multiplicity and angular curvature.

3.5 Ring, overlap, and fixed-anisotropy regimes

Put

$$\alpha := \varepsilon V_4, \quad \omega_c := \omega_0 + \varepsilon V_0, \quad t_c := |\alpha|^{-1}. \quad (136)$$

Three regimes must be kept distinct.

Pre-crossover ring regime. If

$$t_{\text{rad}} \ll t \ll t_c, \quad (137)$$

then $|\tau| \ll 1$ and

$$J_0(\tau) = 1 - \frac{\tau^2}{4} + O(\tau^4). \quad (138)$$

The continuous stationary ring has already localized radially but not angularly, so the nonnodal response has the Morse–Bott law $|G^+| = O(t^{-1/2})$.

Bessel–Morse overlap. For $|\tau| \gg 1$, the large-argument expansion is

$$J_0(\tau) = \sqrt{\frac{2}{\pi|\tau|}} \left[\cos\left(|\tau| - \frac{\pi}{4}\right) + O(|\tau|^{-1}) \right]. \quad (139)$$

The first-order angular phase remains accurate in this late-Bessel regime only while its accumulated higher-order error is small. Let M_3 be a uniform bound such that the reduced-phase remainder after the displayed second-order term is at most $M_3|\varepsilon|^3$ in C^4 . The correct condition is

$$t \gg t_{\text{rad}}, \quad |\varepsilon V_4|t \gg 1, \quad \varepsilon^2 t \|W_2^{\text{eff}}\|_{L^\infty(0,2\pi)} \ll 1, \quad |\varepsilon|^3 t M_3 \ll 1, \quad W_2^{\text{eff}} = W_2 - \frac{B^2}{2a}. \quad (140)$$

When V_4 , W_2^{eff} , and the radial scales are order one, this overlap has the familiar form $|\varepsilon|^{-1} \ll t \ll |\varepsilon|^{-2}$. The compact- τ theorem does not, by itself, justify taking $|\tau| \rightarrow \infty$. The overlap requires a second, growing-parameter stationary-phase argument.

Theorem 3.2 (Growing-parameter Bessel–Morse overlap). *Assume the hypotheses of Theorem 3.1, with the phase and amplitude bounds uniform in a fixed small parameter interval. Let $\varepsilon \rightarrow 0$ and $t \rightarrow \infty$ so that, with $\lambda := |\varepsilon V_4|t$,*

$$\frac{t}{t_{\text{rad}}} \rightarrow \infty, \quad \lambda \rightarrow \infty, \quad \varepsilon^2 t \|W_2^{\text{eff}}\|_{L^\infty} \rightarrow 0, \quad |\varepsilon|^3 t M_3 \rightarrow 0. \quad (141)$$

Then

$$G^+(t; \varepsilon) = \mathcal{C} \sqrt{\frac{2}{\pi|\alpha|}} \frac{e^{-i\omega_c t}}{t} \cos\left(|\alpha|t - \frac{\pi}{4}\right) + \mathcal{E}_{\text{ov}}(t, \varepsilon), \quad (142)$$

where, uniformly in this two-parameter regime,

$$\mathcal{E}_{\text{ov}} = \frac{1}{t\sqrt{|\alpha|}} O(\lambda^{-1} + |\varepsilon| + \varepsilon^2 t \|W_2^{\text{eff}}\|_{L^\infty} + |\varepsilon|^3 t M_3) + O(t^{-3/2}). \quad (143)$$

Consequently Eq. (142) is a relative asymptotic equivalence on subsequences bounded away from the cancellation zeros, for example when $|\cos(\lambda - \pi/4)| \geq c > 0$.

Proof. After removing the carrier in Eq. (120), define

$$\begin{aligned} \Psi_\varepsilon(\theta) &:= \frac{\omega_*(\theta; \varepsilon) - \omega_c}{|\alpha|} = s \cos 4\theta + O_{C^4}(|\varepsilon|), \quad s := \text{sign}(\alpha), \\ \mathcal{B}_\varepsilon(\theta) &:= \frac{(k_0 + q_*)\mathcal{A}(q_*, \theta; \varepsilon)}{\sqrt{\omega_{qq}(q_*, \theta; \varepsilon)}} = \frac{k_0 A_0}{\sqrt{a}} + O_{C^2}(|\varepsilon|). \end{aligned} \quad (144)$$

The C^4 phase estimate follows from the stated C^6 regularity after shrinking the annulus. The normalized perturbation is order $\varepsilon^2/|\alpha| = |\varepsilon|/|V_4|$. Hence the eight simple critical points of $s \cos 4\theta$ persist, exhaust the angular circle, and satisfy

$$\theta_j(\varepsilon) = \frac{j\pi}{4} + O(|\varepsilon|), \quad |\Psi_\varepsilon''(\theta_j)| = 16 + O(|\varepsilon|). \quad (145)$$

One-dimensional stationary phase with large parameter λ , applied directly to this perturbed phase, gives

$$\int_0^{2\pi} \mathcal{B}_\varepsilon e^{-i\lambda \Psi_\varepsilon} d\theta = \left(\frac{2\pi}{\lambda}\right)^{1/2} \sum_{j=0}^7 \frac{\mathcal{B}_\varepsilon(\theta_j) e^{-i\lambda \Psi_\varepsilon(\theta_j) - i\pi \text{sign}(\Psi_\varepsilon''(\theta_j))/4}}{\sqrt{|\Psi_\varepsilon''(\theta_j)|}} + O(\lambda^{-3/2}). \quad (146)$$

Expanding the eight coefficients back to their unperturbed values costs $O(|\varepsilon|)$. Their accumulated phase correction is $O(\varepsilon^2 t \|W_2^{\text{eff}}\|_{L^\infty} + |\varepsilon|^3 t M_3)$, as required in Eq. (141). Four angular curvatures are positive and four are negative. Multiplying their one-dimensional signature factors by the radial factor $e^{-i\pi/4}$ produces, respectively, the minimum factor $-i$ and the saddle factor 1 . Summing the two fourfold families gives the leading term in Eq. (142); the angular remainder and the radial $O(t^{-3/2})$ remainder give Eq. (143). This proof is independent of the compact- τ limit. $\square$

Away from the cancellation zeros, the envelope is $O(|\alpha|^{-1/2}t^{-1})$. Pointwise logarithmic slopes through the zeros are undefined; an upper, RMS, or carrier-demodulated envelope must be used with a declared fitting window.

Bessel constants as a Morse sum. The agreement is stronger than equality of exponents. Consider the general first-order phase

$$\omega = \omega_c + \frac{a}{2}q^2 + \alpha \cos[m(\theta - \theta_0)], \quad m \in 2\mathbb{N}, \quad \alpha \neq 0. \quad (147)$$

There are m points at the lower frequency $\omega_- = \omega_c - |\alpha|$ and m at the upper frequency $\omega_+ = \omega_c + |\alpha|$. Since $a > 0$, the lower points are minima and the upper points are saddles. To leading order their Cartesian Hessians satisfy

$$\sqrt{|\det H_j|} = \frac{m\sqrt{a|\alpha|}}{k_0}, \quad \text{sig}(H_-) = 2, \quad \text{sig}(H_+) = 0. \quad (148)$$

The m -fold multiplicity of each family cancels the factor m in Eq. (148). For constant amplitude A_0 , ordinary two-dimensional stationary phase therefore gives

$$G_{\text{Morse}}^+ \sim \frac{\mathcal{F}_2(2\pi)k_0A_0}{t\sqrt{a|\alpha|}} \left[e^{-i\omega_-t - i\pi/2} + e^{-i\omega_+t} \right]. \quad (149)$$

On the other hand, substituting Eq. (128) into Eq. (142) and expanding the cosine gives

$$\begin{aligned} G_{\text{Bessel}}^+ &\sim \frac{\mathcal{F}_2(2\pi)k_0A_0}{t\sqrt{a|\alpha|}} \left[e^{-i\pi/4} e^{-i\omega_c t} e^{i(|\alpha|t - \pi/4)} + e^{-i\pi/4} e^{-i\omega_c t} e^{-i(|\alpha|t - \pi/4)} \right] \\ &= \frac{\mathcal{F}_2(2\pi)k_0A_0}{t\sqrt{a|\alpha|}} \left[e^{-i\omega_-t - i\pi/2} + e^{-i\omega_+t} \right]. \end{aligned} \quad (150)$$

Thus the Fourier constant, both frequencies, the minimum and saddle signature phases, and both multiplicities match exactly. For the cubic case $m = 4$, this is the equality between four minima plus four saddles and the large-argument asymptotics of the single J_0 kernel.

3.6 Exact Cartesian Morse theorem at fixed anisotropy

The overlap calculation still uses the first-order phase. The ultimate fixed-anisotropy result must instead use the exact branch frequency, exact critical points, and exact Cartesian Hessians.

Theorem 3.3 (Fixed-anisotropy Morse decay). *Fix a sufficiently small $\varepsilon \neq 0$. Suppose the exact selected branch $\omega(\mathbf{k}; \varepsilon)$ and amplitude $\mathcal{A}(\mathbf{k}; \varepsilon)$ are smooth on the support of the local window, every stationary point $\mathbf{k}_j$ there is isolated and nondegenerate, and no stationary point lies on the support boundary. Let*

$$\omega_j := \omega(\mathbf{k}_j; \varepsilon), \quad H_j := \nabla_{\mathbf{k}}^2 \omega(\mathbf{k}_j; \varepsilon), \quad \sigma_j := \text{sig}(H_j) = n_+(H_j) - n_-(H_j), \quad \mathcal{A}_j := \mathcal{A}(\mathbf{k}_j; \varepsilon). \quad (151)$$

Then, as $t \rightarrow \infty$ at this fixed ε ,

$$G^+(t; \varepsilon) = \frac{2\pi\mathcal{F}_2}{t} \sum_j \frac{\mathcal{A}_j e^{-it\omega_j - i\pi\sigma_j/4}}{\sqrt{|\det H_j|}} + O_\varepsilon(t^{-2}). \quad (152)$$

Hence a generic nonnodal isolated critical point contributes t^{-1} . The constant in the remainder may diverge as $\varepsilon \rightarrow 0$, so Eq. (152) is deliberately not a uniform coalescing-critical-point formula. The physical real response is again twice the real part of the right-hand side.

Proof and polar-Jacobian cancellation. Apply the two-dimensional stationary-phase theorem in Cartesian wave-vector coordinates to Eq. (119). For the phase $-\omega$, the signature factor is $\exp[-i\pi \text{sig}(H_j)/4]$, and the Cartesian Fourier measure is simply $\mathcal{F}_2 d^2\mathbf{k}$. This directly gives Eq. (152).

The same result in polar coordinates makes an important cancellation explicit. At a critical point, let

$$H_j^{(q,\theta)} := \begin{pmatrix} \omega_{qq} & \omega_{q\theta} \\ \omega_{q\theta} & \omega_{\theta\theta} \end{pmatrix}_{(q_j, \theta_j)}. \quad (153)$$

Equation (87) implies

$$\det H_j^{(q,\theta)} = k_j^2 \det H_j, \quad \frac{k_j \mathcal{A}_j}{\sqrt{|\det H_j^{(q,\theta)}|}} = \frac{\mathcal{A}_j}{\sqrt{|\det H_j|}}, \quad k_j > 0. \quad (154)$$

Thus the polar factor k_j cancels against the Hessian transformation. The amplitude $\mathcal{A}_j$ in the Cartesian Morse sum contains the modal projection, Gaussian source, and branch window, but it does not contain a residual polar Jacobian. A minimum has signature $+2$, a saddle has signature 0 , and a maximum has signature -2 , giving the factors $-i$, 1 , and $+i$, respectively. This completes the proof. $\square$

3.7 Frequency separation and signed modulation

For the exact fourfold branch, symmetry groups the eight points into a fourfold minimum family and a fourfold saddle family. Define their exact frequencies by $\omega_{\min}(\varepsilon)$ and $\omega_{\text{sad}}(\varepsilon)$. Their exact critical-frequency separation and its weak-anisotropy expansion are

$$\Delta\omega_{\text{ms}}^{\text{exact}} := |\omega_{\text{sad}} - \omega_{\min}|, \quad \Delta\omega_{\text{ms}}^{\text{exact}} = 2|\varepsilon V_4| + O(\varepsilon^2). \quad (155)$$

Factoring a mean carrier from two equal-weight leading terms produces a signed modulation whose angular rate is half the separation:

$$\Omega_{\text{mod}} = \frac{\Delta\omega_{\text{ms}}^{\text{exact}}}{2} = |\varepsilon V_4| + O(\varepsilon^2). \quad (156)$$

Here $\Omega_{\text{mod}} \geq 0$ is the magnitude of the angular rate of a signed oscillatory modulation, not a signed frequency. In the first-order overlap its signed-cosine period and magnitude/RMS repetition period are

$$T_{\text{signed}} = \frac{2\pi}{\Omega_{\text{mod}}}, \quad T_{|\cdot|} = \frac{\pi}{\Omega_{\text{mod}}} = \frac{2\pi}{\Delta\omega_{\text{ms}}^{\text{exact}}}. \quad (157)$$

In the first-order overlap the signature phases make the signed factor precisely $\cos(|\varepsilon V_4|t - \pi/4)$, as shown in Eq. (142). Its magnitude repeats twice during one signed-cosine period. Therefore the critical-frequency separation and the signed modulation rate differ by a factor of two. They should be reported separately rather than combined under an ambiguous lossless continuum “beat frequency.” At fixed ε , unequal exact amplitudes or higher-order phase shifts generally turn the pure cosine into a two-phaser modulation without changing the exact separation.

3.8 Nodal channels, noncommuting limits, and theorem boundaries

The nonzero-amplitude hypotheses are essential rather than cosmetic. There are three distinct nodal mechanisms.

1. If the isotropic angular mean A_0 vanishes, the canonical J_0 term is absent. Equation (134) supplies the correct leading Bessel order when another compatible Fourier coefficient is nonzero.
2. If $\mathcal{A}_j = 0$ at an exact Morse point, that point has no t^{-1} contribution in Eq. (152). The first nonzero jet of the amplitude at the point determines the next stationary-phase term; its exponent and phase must be derived for that source/detector channel.
3. Even when every local amplitude is nonzero, symmetry-related terms can cancel at discrete times or in a selected detector combination. Such zeros do not change the generic local stationary-phase power.

These cases also explain why response-decay fits use a predeclared RMS, upper, or demodulated envelope and exclude neighborhoods of cancellation zeros.

Finally, the two relevant limits are noncommuting limits:

$$\begin{aligned} \varepsilon \rightarrow 0 \text{ before the ultimate-time limit} &\implies \text{a Morse–Bott ring and } t^{-1/2}, \\ t \rightarrow \infty \text{ at fixed } \varepsilon \neq 0 &\implies \text{isolated exact Morse points and } t^{-1}. \end{aligned} \quad (158)$$

Indeed, $t_c = |\varepsilon V_4|^{-1} \rightarrow \infty$ as $\varepsilon \rightarrow 0$, whereas the coefficient of the separated Morse sum scales as $|\varepsilon V_4|^{-1/2}$ and becomes nonuniform when the points merge into a ring. The uniform J_0 theorem resolves this coalescence for $\varepsilon t = O(1)$; it cannot be extrapolated to arbitrarily large time at fixed nonzero ε , because then $\varepsilon^2 t W_2^{\text{eff}}$ eventually becomes order one.

The results of this section require a finite-radius positive-frequency ring, nonzero radial curvature, a simple smooth branch with a uniform local eigengap, a stationary set separated from the window boundary, and exact nondegenerate Hessians in the fixed-anisotropy theorem. If $a = 0$, if $V_4 = 0$ without a resolved higher harmonic, if the eigengap closes, or if an exact Hessian is singular, a different canonical scaling is required. Material damping introduces an additional exponential envelope and lies outside the present lossless power-law theorem. These boundaries prevent either Eq. (127) or Eq. (152) from being applied beyond its proved regime.

4 Thickness-resolved spectral numerics and error control

This section fixes the discrete weak form, branch tracking, perturbation solves, critical-point resolution checks, response quadrature, and acceptance rules used by every numerical artifact. The objective is reproducibility: each symbol is tied below to a code interface, a qualified configuration field, or a qualified artifact field.

4.1 GLL element map and traction-free weak form

We use the same nondimensionalization as in Sec. 1: h is the half-thickness, $z \in [-h, h]$, $k = hk^{\text{phys}}$, and $\omega = h\omega^{\text{phys}}/c_T$. A separated displacement has the form

$$\mathbf{u}(x, y, z, t) = \mathbf{U}(z) \exp\{i(k_x x + k_y y) - i\omega t\}.$$

On element $e = [z_e^-, z_e^+]$, let p be the polynomial degree and $\{\xi_j, w_j\}_{j=0}^p$ the $p+1$ Legendre–Gauss–Lobatto (GLL) nodes and weights returned by `gll_nodes_weights`. The physical map, Jacobian, weights, and derivative matrix are

$$\begin{aligned} z_e(\xi) &= \frac{z_e^- + z_e^+}{2} + \frac{z_e^+ - z_e^-}{2}\xi, & -1 \leq \xi \leq 1, \\ J_e &= \frac{z_e^+ - z_e^-}{2}, & w_{ej} = J_e w_j, & D_e = J_e^{-1} D. \end{aligned} \quad (159)$$

Here D is the barycentric matrix produced by `differentiation_matrix`.

The routine `build_gll_mesh` shares the interface node of adjacent elements and maps the local connectivities to node-major triples (U_x, U_y, U_z) . Thus one element on $[-h, h]$ is the production discretization for a homogeneous plate, while a split mesh with bounds $(-h, 0, h)$ supplies an independent assembly check.

For test function $\mathbf{V}$, integration by parts in z gives

$$\int_{-h}^h \boldsymbol{\varepsilon}(\mathbf{V})^* : \mathbf{C} : \boldsymbol{\varepsilon}(\mathbf{U}) dz = \omega^2 \int_{-h}^h \rho \mathbf{V}^* \cdot \mathbf{U} dz + \sum_{z \in \{-h, h\}} \mathbf{V}(z)^* \cdot [\boldsymbol{\sigma}(z) \mathbf{n}(z)]. \quad (160)$$

No displacement degree of freedom is constrained at either face. The boundary term vanishes precisely when

$$\boldsymbol{\sigma} \mathbf{n}|_{z=\pm h} = 0 \quad \text{is the natural traction-free boundary condition.} \quad (161)$$

$$\boldsymbol{\varepsilon}_M = (\varepsilon_{xx}, \varepsilon_{yy}, \varepsilon_{zz}, \sqrt{2}\varepsilon_{yz}, \sqrt{2}\varepsilon_{xz}, \sqrt{2}\varepsilon_{xy})^\top. \quad (162)$$

This Mandel convention makes the Euclidean product equal to the tensor contraction, $\boldsymbol{\varepsilon}_M^* \mathbf{C}_M \boldsymbol{\varepsilon}_M$. With nodal interpolant $\mathbf{N}_j$, the strain matrix at a quadrature node is $\mathbf{B} = \mathbf{B}_0 + k_x \mathbf{B}_x + k_y \mathbf{B}_y$. The element and assembled matrices used by `assemble_plate_matrices` are

$$\begin{aligned} \mathbf{K}_e &= \sum_{j=0}^p w_{ej} \mathbf{B}_j^* \mathbf{C}_M \mathbf{B}_j, \\ \mathbf{M}_e &= \sum_{j=0}^p w_{ej} \rho \mathbf{N}_j^* \mathbf{N}_j, \\ \mathbf{K} &= \text{assemble } \mathbf{K}_e, & \mathbf{M} &= \text{assemble } \mathbf{M}_e. \end{aligned}$$

Consistent collocated GLL quadrature is used for stiffness and mass. Because the cardinal basis is evaluated at the same GLL nodes, the assembled mass is diagonal. This is quadrature-induced mass lumping; no separate post-assembly lumping or penalty term is applied. The reported Hermitian defect is $\|\mathbf{K} - \mathbf{K}^*\|_F / \|\mathbf{K}\|_F$. Constitutive input is rejected unless its minor and major symmetries hold and its Mandel matrix is positive definite.

4.2 Matrix derivatives and eigensystem diagnostics

Because $\mathbf{B}$ is affine in (k_x, k_y) , the five derivative matrices are assembled analytically at the same quadrature nodes by the wavevector-derivative assembler:

$$\begin{aligned} \mathbf{K}_i &= \sum_{e,j} w_{ej} (\mathbf{B}_i^* \mathbf{C}_M \mathbf{B} + \mathbf{B}^* \mathbf{C}_M \mathbf{B}_i), \\ \mathbf{K}_{ii} &= 2 \sum_{e,j} w_{ej} \mathbf{B}_i^* \mathbf{C}_M \mathbf{B}_i, \\ \mathbf{K}_{xy} &= \sum_{e,j} w_{ej} (\mathbf{B}_x^* \mathbf{C}_M \mathbf{B}_y + \mathbf{B}_y^* \mathbf{C}_M \mathbf{B}_x), & i \in \{x, y\}, & \partial_{k_i} \mathbf{M} = 0. \end{aligned} \quad (163)$$

The generalized Hermitian problem $\mathbf{K}\mathbf{u}_n = \lambda_n \mathbf{M}\mathbf{u}_n$, $\lambda_n = \omega_n^2$, is solved by `solve_plate_modes`. The solver uses SciPy's generalized Hermitian `gvx` driver for the lowest twelve modes and requires a positive-definite mass. It rejects $\lambda_{\min} < -\tau_-$, while values in $[-\tau_-, 0]$ are clipped to zero, where

$$\tau_- = 10^3 \epsilon_{\text{mach}} \max\left(\frac{\|\mathbf{K}\|_2}{\lambda_{\min}(\mathbf{M})}, \text{tiny}_{64}\right).$$

Every eigenvector is explicitly mass normalized. `tiny64` denotes the smallest positive normal binary64 number. For the selected mode the code's normalized residual is

$$r = \frac{\|\mathbf{K}\mathbf{u} - \lambda \mathbf{M}\mathbf{u}\|_2}{\max\{\|\mathbf{K}\mathbf{u}\|_2 + \|\lambda \mathbf{M}\mathbf{u}\|_2, \text{tiny}_{64}\}} \quad (\text{normalized residual}). \quad (164)$$

$$d_M = \|\mathbf{U}^* \mathbf{M} \mathbf{U} - \mathbf{I}\|_F \quad (\text{mass orthogonality defect}). \quad (165)$$

$$\text{MAC}_M(u, v) = \frac{|u^* \mathbf{M} v|^2}{(u^* \mathbf{M} u)(v^* \mathbf{M} v)}. \quad (166)$$

Mode tracking. For two nonzero vectors, `mass_mac` evaluates Eq. (166). The continuation rule is

predict the eigenvalue and maximize the previous-subspace projection; form the connected close-eigenvalue cluster and test its overlap; require scalar mass MAC or squared cluster overlap, then phase align; retain both the selected vector and the complete cluster basis.	(167)
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If $s_{\min}$ is the smallest principal cosine between the previous and current mass-orthonormal cluster bases, a step with threshold η passes when

$$\text{MAC}_M \geq \eta \quad \text{or} \quad s_{\min}^2 \geq \eta.$$

The scalar/squared-subspace threshold is $\eta = 0.2$ in the generic tracker and $\eta = 0.8$ in the registered robustness sweeps. The connected-cluster relative eigenvalue threshold is 10^{-3} .

`RingAnchoredSpectralEvaluator` constructs eight endpoint-free angular anchors independently at orders p and $p + 4$, requires angular closure scalar MAC and the unsquared principal cosine each to be at least $1 - 10^{-8}$, then continues monotonically away from $q = 0$ on separate inward and outward rays. It also rejects a cluster touching the top of the twelve computed modes. The relative eigengap returned by the tracker is

$$g_{\text{rel}} = \min_{m \neq n} \frac{|\lambda_m - \lambda_n|}{\max(|\lambda_n|, |\lambda_m|, 1)}, \quad (168)$$

$$g_{\text{rel}} > 10 \delta \lambda_{\text{rel}}.$$

A sample that fails the second line is rejected by the eigengap gate.

Here

$$\delta \lambda_{\text{rel}} = \max\left\{\frac{|\lambda^{(p+4)} - \lambda^{(p)}|}{\max(|\lambda^{(p+4)}|, |\lambda^{(p)}|, 1)}, r^{(p)}, r^{(p+4)}\right\}.$$

The same nested-order pair defines absolute frequency and gradient discrepancy estimators as the differences between the two tracked answers. These inter-order estimators quantify resolution sensitivity; without a continuum enclosure or a posteriori theorem, they are not upper bounds on the unknown continuum errors. This gate is essential: every derivative formula below assumes a simple, spectrally isolated branch.

4.3 Sensitivities and critical-point resolution checks

Let r denote the radial wavenumber and let ϵ multiply the cubic constitutive perturbation at fixed density. At unit mass norm, $\lambda_p = u^* \mathbf{K}_p u$. The radial mode derivative is found by the bordered reduced-resolvent system implemented in `radial_frequency_sensitivity`:

$$(\mathbf{K} - \lambda \mathbf{M})u_p = -(\mathbf{K}_p - \lambda_p \mathbf{M})u, \quad u^* \mathbf{M} u_p = 0, \quad \lambda_p = u^* \mathbf{K}_p u. \quad (169)$$

Row and constraint scalings are applied before the dense solve; the original equation and gauge are then checked with relative residual 10^{-8} . The mixed sensitivity includes both the direct matrix derivative and the mode derivative,

$$\lambda_{r\epsilon} = u^* \mathbf{K}_{r\epsilon} u + 2 \operatorname{Re}(u^* \mathbf{K}_\epsilon u_r), \quad V = \frac{\lambda_\epsilon}{2\omega}, \quad B = \frac{\lambda_{r\epsilon}}{2\omega} - \frac{\lambda_\epsilon \lambda_r}{4\omega^3}.$$

For the primary sensitivity artifact, the matrix perturbations are constructed at each angle from the exactly linear cubic family as

$$\mathbf{K}_\epsilon = \frac{\mathbf{K}(+0.25) - \mathbf{K}(-0.25)}{0.5}, \quad \mathbf{K}_{r\epsilon} = \frac{\mathbf{K}_r(+0.25) - \mathbf{K}_r(-0.25)}{0.5}.$$

The full profile uses order 10, 64 endpoint-free angles, and a single centered physical check with $\Delta\epsilon = \Delta q = 0.0025$; the smoke profile uses order 8, 32 angles, and 0.005. On an N_θ -point grid the reported cubic coefficient is the discrete Fourier coefficient

$$V_4 = \frac{2}{N_\theta} \sum_{j=0}^{N_\theta-1} V(\theta_j) \cos(4\theta_j),$$

and $V_0 = N_\theta^{-1} \sum_j V(\theta_j)$. The primary artifact accepts only when the relative vector errors of both V and B , compared with that single centered full-wave check, are below the configured field

`config/reference.yaml::sensitivity_match_tolerance.`

The separate convergence artifact supplies step-refinement evidence. It uses the tensor-product centered grid $\Delta\epsilon, \Delta q \in \{0.02, 0.01, 0.005, 0.0025\}$, checks the last two rows/columns against the same tolerance, and requires improvement relative to the coarsest row/column. Thus only this independent sweep varies both step sizes; it does not differentiate the perturbation implementation against itself.

seed every sign-changing polar cell; solve $\nabla_{\mathbf{k}} \omega = 0$ and deduplicate; classify the Cartesian Hessian; check both annular boundaries and the index sum.	(170)
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The production implementation is the registered critical-point locator. Its search annulus is

$$A = \{\mathbf{k} : (1 - f_A)k_0 \leq |\mathbf{k}| \leq (1 + f_A)k_0\}.$$

Here f_A is `config/reference.yaml::annulus_fraction`. On a polar candidate grid, the Cartesian gradient is resolved into radial and tangential components. Every cell across which both components bracket zero supplies a seed. SciPy's `scipy.optimize.root` follows each seed with its default Powell hybrid method (`method=hybr`). The finite-difference Cartesian Hessian is supplied as the Jacobian, with solver tolerance 10^{-11} . Candidates outside A or with gradient residual larger than their nested-order gradient-discrepancy estimator are rejected. An unresolved Hessian is not silently discarded: classification raises an error and fails the resolution-check stage. Remaining roots are merged only when their separation does not exceed d_{merge} , where

$$\begin{aligned} b_0 &= 5 \times 10^{-9} \max(k_0, 1), \\ b_{\max} &= \max\{b_0, \min(0.05 \Delta_A, 0.01 k_0)\}, \\ b_{12} &= \max\{b_0, \delta k_{\text{lin},1} + \delta k_{\text{lin},2}\}, \\ d_{\text{merge}} &= \min(b_{12}, b_{\max}), \end{aligned}$$

where Δ_A is the annular half-width.

The full workflow uses a 5×16 candidate grid and independently repeats the search on 9×32 . Counts, kinds, positions, frequencies, and Hessian eigenvalues must agree within the following propagated bounds. For matched coarse/fine roots, define

$$\begin{aligned} d_k &= \|\mathbf{k}_c - \mathbf{k}_f\|_2, \\ u_k &= \frac{\delta g_c}{\min_\ell |\lambda_\ell(H_c)|} + \frac{\delta g_f}{\min_\ell |\lambda_\ell(H_f)|}, \\ H_{\max} &= \max_{\nu \in \{c,f\}} \max_\ell |\lambda_\ell(H_\nu)|, \\ d_H &= \max_\ell |\lambda_\ell(H_c) - \lambda_\ell(H_f)|. \end{aligned}$$

The comparison passes only if

$$\begin{aligned} d_k &\leq \max(5 \times 10^{-9} k_0, u_k), \\ |\omega_c - \omega_f| &\leq \delta\omega_c + \delta\omega_f + \frac{1}{2} H_{\max} d_k^2, \\ d_H &\leq \delta H_c + \delta H_f. \end{aligned}$$

These candidate-grid bounds are distinct from the capped deduplication radius used within either search. The full critical-point profile uses Hessian step 10^{-3} ; the smoke profile uses 2×10^{-3} . Both use $N_b = 32$ and $2N_b = 64$ boundary samples for the final annular consistency check. The Green workflow repeats the accepted search on the 15×32 response-support domain specified below; the silicon workflow uses 9×32 . Both use boundary levels 16/32; their Hessian steps are 10^{-3} (full Green) and 3×10^{-3} (silicon), respectively. The negative- ϵ search is additionally rotated by $\pi/16$, so its success is not caused by placing candidate cells on the positive- ϵ symmetry rays. The recorded position scale follows from the linearized gradient equation:

$$\delta k_{\text{lin}} := \frac{\delta g}{\min_{\ell} |\lambda_{\ell}(H)|} = \|H^{-1}\|_2 \delta g, \quad H = \nabla_{\mathbf{k}}^2 \omega. \quad (171)$$

Equation (171) is a local resolution scale, not a nonlinear interval enclosure. Its interpretation requires a nondegenerate Hessian that remains stable over the displacement. Here δg is the code field `gradient_uncertainty`. The root residual is checked separately and must not exceed δg ; a conservative first-order displacement estimate under the linearized Hessian would replace δg by $\delta g + \|\nabla_{\mathbf{k}} \omega\|_2$.

The Hessian is a Richardson extrapolation of centered differences of the full-wave gradient. Its uncertainty is the fine-versus-Richardson difference plus the two nested-order gradient uncertainties propagated through the stencil. Classification is accepted only if the smallest absolute Hessian eigenvalue exceeds ten times this uncertainty. A minimum or maximum has gradient index +1, whereas a saddle has gradient index -1. Despite the legacy artifact field name `critical_points.morse_index`, it stores this Poincaré gradient index, not the Morse index (the number of negative Hessian eigenvalues). The pointwise `gradient_uncertainty` drives acceptance and uncertainty propagation but is not stored as a critical-point array; the sidecar stores the boundary maxima and the artifact stores `critical_points.gradient_residual`.

$$\text{wind}_{\partial A}(\nabla_{\mathbf{k}} \omega) = \sum_{k_j \in A} \text{ind}(k_j), \quad \text{and the boundary winding index closes.} \quad (172)$$

The independent routine `verify_annular_exhaustion` samples both annular boundaries with N_b and $2N_b$ endpoint-free angles. It requires (i) the minimum boundary-gradient norm to exceed ten times the maximum gradient uncertainty, (ii) equal coarse and fine boundary winding, (iii) every fine angular increment of the gradient direction to be less than $\pi/2$, and (iv) the outer-minus-inner boundary winding to equal the sum of accepted Poincaré indices. Thus the boundary winding and the root index closes on both resolutions. This is a finite-resolution boundary/index consistency check: by itself it cannot exclude an unresolved, mutually canceling critical-point pair in the annular interior. Candidate-grid doubling, symmetry-offset repetition, and Hessian separation provide the additional acceptance checks used here. The perturbative eight-point prediction is stored for comparison, but its prediction error is not a root-acceptance gate.

4.4 Polar Green quadrature, fit masks, and phase accuracy

The endpoint-free grid is $\theta_j = 2\pi j/N_{\theta}$. The production Green workflow exploits verified C_{4v} covariance: its `_CachedCubicSymmetryFactory` maps every angle to

$$r_{\theta} = \theta \bmod \frac{\pi}{2}, \quad \theta_{\text{can}} = \min\left(r_{\theta}, \frac{\pi}{2} - r_{\theta}\right)$$

and rounds θ_{can} and the radial cache key to 14 decimal places. It reuses samples at equal $(\theta_{\text{can}}, \text{sign } q, k_0 + q)$. Each canonical ray starts fresh inward and outward ring-anchored spectral continuations at $q = 0$. Symmetry-equivalent rays therefore reuse the canonical full-wave value; they are not recomputed as independent solves. Every refinement level owns a separate cache, so no coarse-grid result is used as a fine-grid answer.

At a surface node the phase-invariant normal-impulse modal factor is

$$A(q, \theta) = \frac{i}{2\omega(q, \theta)} |u_z(z = h; q, \theta)|^2.$$

Let $A^{(p)}$ and $A^{(p+4)}$ denote the two spectral-order amplitudes. The full-wave radial-evaluator factory requires

$$|A^{(p+4)} - A^{(p)}| \leq 10^{-12} + 10^{-2} \max(|A^{(p+4)}|, |A^{(p)}|).$$

At $q = 0$, independently created inward and outward anchors must also agree:

$$\begin{aligned} |\omega_{\text{in}} - \omega_{\text{out}}| &\leq \delta\omega_{\text{in}} + \delta\omega_{\text{out}} + 32\epsilon_{\text{mach}} \max(\omega_{\text{in}}, \omega_{\text{out}}, 1), \\ |A_{\text{in}} - A_{\text{out}}| &\leq \delta A_{\text{in}} + \delta A_{\text{out}} + 32\epsilon_{\text{mach}} \max(|A_{\text{in}}|, |A_{\text{out}}|, 1). \end{aligned}$$

The Gaussian source and the local super-Gaussian branch window are fixed, not fitted:

$$S(k) = \exp[-(Rk/2)^2], \quad W(q) = \exp[-(q/\sigma)^8]b_\sigma(q).$$

The ratio R/h is mapped to

$$\text{config/reference.yaml}::\text{source_radius_over_h}.$$

The ratio σ/k_0 is mapped to

$$\text{config/reference.yaml}::\text{window_sigma_over_k0}.$$

The deterministic C^∞ taper b_σ equals one for $|q| \leq 1.25\sigma$, follows the standard flat $\exp(-1/x)$ transition, and vanishes identically for $|q| \geq 1.5\sigma$. The radial interval $[-1.5\sigma, 1.5\sigma]$ therefore contains the complete support, contains exactly one $q = 0$ node, keeps $k_0 + q > 0$, and has no endpoint term. The endpoint-free trapezoidal rule in θ and composite Simpson rule in q evaluate the Cartesian Fourier measure without using a Bessel approximation:

$$\begin{aligned} F_{\ell j} &= (k_0 + q_\ell)S(k_0 + q_\ell)W(q_\ell)A_{\ell j}, \\ G_h(t) &= \frac{\text{Simpson}_{q_\ell}}{(2\pi)^2} \left[\frac{2\pi}{N_\theta} \sum_{j=0}^{N_\theta-1} F_{\ell j} e^{-i\omega_{\ell j}t} \right], \\ \tilde{G}_h(t) &= e^{i\omega_c t} G_h(t), \quad \omega_c = \omega_0 + \epsilon V_0. \end{aligned} \tag{173}$$

The registered-grid integration interface exposes the generic three-level construction. The production Green workflow rebuilds each level as an independent full-wave tracked surface, while retaining exact nested nodes for discrepancy evaluation. It uses two distinct registered three-grid sequences:

$$\begin{array}{l|lll} \text{uniform-crossover rows } (\epsilon \neq 0.08) & 129 \times 32 & 257 \times 64 & 513 \times 128 \\ \text{fixed-anisotropy row } (\epsilon = 0.08) & 129 \times 64 & 257 \times 128 & 513 \times 256 \end{array}$$

The fixed-Morse comparison searches the tracked branch on $|q| \leq 1.55\sigma = 0.2325k_0$, which strictly contains the direct support $|q| \leq 1.5\sigma = 0.225k_0$ with margin 0.05σ . It uses a 15×32 candidate grid and $16/32$ -node boundary/index checks. Every resolved stationary point in this search domain is multiplied by the same b_σ and included in the Morse sum. The sidecar records the two domains, margin, boundary checks, point counts, and contribution counts for each ϵ .

The dimensions are radial-by-angular. Separate caches are created at every level, so refinement does not reuse a coarse numerical frequency as a fine answer. By contrast, the full-profile convergence artifact's auxiliary robustness study constructs a 513×128 reference surface and uses nested subsampling to 65×16 , 129×32 , and 257×64 . The smoke profile uses a 257×128 reference and 33×16 , 65×32 , and 129×64 subsamples. This auxiliary study measures quadrature and interpolation convergence; it is not substituted for the independent full-wave production grids.

The auxiliary responses use the same 61 times $t_m = 1500 + 75m$, $0 \leq m \leq 60$. If G_ℓ is the response from subsampled level ℓ and G_{ref} is the reference response, then

$$\|f\|_{t,\text{RMS}} = \left[M_t^{-1} \sum_m |f(t_m)|^2 \right]^{1/2}, \quad Q_\ell = \frac{\|G_\ell - G_{\text{ref}}\|_{t,\text{RMS}}}{\|G_{\text{ref}}\|_{t,\text{RMS}}}.$$

The stored Q_ℓ field is **convergence.quadrature.error**. To measure frequency interpolation, each level first receives a periodic cubic spline in θ , evaluated at every reference angle; an ordinary cubic spline in q is then evaluated at every reference radial node. Writing the resulting surface as $\mathcal{I}_q \mathcal{I}_\theta \omega_\ell$, the stored error is

$$\begin{aligned} I_\ell &= \max_{(q,\theta) \in \mathcal{G}_{\text{ref}}} |e_\ell(q,\theta)|, \\ e_\ell &= \mathcal{I}_q \mathcal{I}_\theta \omega_\ell - \omega_{\text{ref}}. \end{aligned}$$

The stored field is **convergence.interpolation.error**. Thus the angular-periodic interpolation precedes the radial interpolation; the order is part of the method.

Let G_0, G_1, G_2 be the demodulated responses on three successive production grids. Their RMS complex differences $E_j = \|G_{j+1} - G_j\|_{\text{RMS}}$ must decrease unless the fine discrepancy is below the roundoff floor

$$E_{\text{round}} = 64\epsilon_{\text{mach}} \max(\|G_2\|_{\text{RMS}}, 1).$$

The final discrepancy must satisfy

$$E_1 \leq 10^{-8} + 0.05\|G_2\|_{\text{RMS}}, \quad E_1/E_0 \leq 0.5$$

whenever the rate lies above the absolute/roundoff floor. The auxiliary surface study separately requires decreasing quadrature and interpolation errors, refinement ratios at most 0.5, and a finest relative quadrature error below 0.05.

The decay exponents are extracted only from preregistered masks. With $\tau = \epsilon V_4 t$,

$$\begin{aligned} M_{\text{early}}(\epsilon, t) &= \mathbf{1}_{\{\epsilon = \epsilon_{\text{first}}\}} \mathbf{1}_{\{t \geq 1500\}} \mathbf{1}_{\{0.10 \leq |\tau| \leq 0.30\}}, \\ M_{\text{late}}(\epsilon, t) &= \mathbf{1}_{\{\epsilon = 0.08\}} \sum_b \mathbf{1}_{[t_b, t_b + T_{\text{ms}})}(t), \\ T_{\text{ms}} &= 2\pi/|\omega_{\text{saddle}} - \omega_{\text{min}}|, \quad t_0 = 1500. \end{aligned} \tag{174}$$

These two arrays are stored as `green_crossover.fit_window_early` and `green_crossover.fit_window_late`. Here $\epsilon_{\text{first}} = 0.005$ in the committed full profile and $\epsilon_{\text{first}} = 0.01$ in the smoke profile; operationally the code uses `row_index == 0`. The early mask must contain at least ten positive-amplitude samples. The call `fit_power_law` performs ordinary least squares on $(\log t, \log |G|)$ and fits both the diagnostic decay slope and a nuisance log-log intercept. That intercept is used only for slope estimation. The separate full-wave/asymptotic comparison applies no fitted alignment of amplitude, phase, carrier frequency, or time. For the late mask, each beat bin is half-open, contains at least three time samples, and is retained only when the complete interval lies in the computed time axis. The fit uses at least four such bins, their geometric time centers, and the direct-response RMS in each bin. The independent fixed-Morse comparison uses $1500 \leq t \leq 10200$, excluding only samples whose exact-Morse phasor coherence is below `_MORSE_COHERENCE_THRESHOLD = 0.30`. Uniform collapse is checked only for $\epsilon \leq 0.04$, $t \geq 1500$, and $|\tau| \leq \text{_UNIFORM_TAU_MAXIMUM} = 2.0$; the $\epsilon = 0.08$ fixed-Morse row is excluded from that first-order gate. The fixed-Morse RMS and maximum discrepancies are normalized by the exact-Morse RMS over the retained noncancellation mask; the cancellation-region RMS uses the same denominator. The uniform-collapse diagnostic is the maximum complex absolute error between `green_crossover.scaled_response` and `green_crossover.J0` on its registered mask.

The operational crossover is the first downward crossing of scaled-response magnitude 0.9, linearly interpolated between adjacent time samples within $[0.5t_{\text{pred}}, 1.5t_{\text{pred}}]$. Absence of such a crossing fails the stage. Because the search bracket is centred on the theoretical $t_{\text{pred}} \propto |\epsilon V_4|^{-1}$, the fitted log slope of crossover time versus ϵ is a theory-centred consistency diagnostic rather than independent evidence for inverse-rate scaling. The complex Bessel collapse is the primary numerical test.

Two production accumulated inter-resolution phase-discrepancy estimators are enforced as acceptance tests. The direct full-wave estimator combines (i) each surface's nested spectral order- p /order- $p + 4$ frequency difference and (ii) the common-node frequency difference between adjacent polar grids. The exact-Morse estimator uses the largest nested-order frequency difference among its eight resolved critical-point contributions. Both call `assert_phase_accuracy`, which enforces

$$t_{\text{max}} \max |\delta\omega| \leq 0.05. \tag{175}$$

The right-hand side is the configured field

$$\text{config/reference.yaml}::\text{phase_error_tolerance}.$$

For a grid pair the $\max |\delta\omega|$ in Eq. (175) is the maximum of its common-node discrepancy and the two adjacent surfaces' largest local spectral discrepancies. The stored `green_crossover.phase_error` is the larger of the direct-grid and exact-Morse accumulated discrepancy estimators. Its legacy field name does not turn this quantity into a continuum error bound.

The auxiliary convergence artifact uses a distinct interpolation discrepancy estimator,

$$\Delta\phi_\ell = t_{\text{max}} (\delta\omega_{\text{interp},\ell} + \max \delta\omega_{\text{local},\ell} + \max \delta\omega_{\text{local},\text{ref}}).$$

Only its finest level is gated against the same 0.05 budget. This diagnostic is not substituted for either production estimator. In short, the declared comparison has no fitted alignment and no response-derived fitting window.

4.5 Declared configuration and fixed acceptance thresholds

The following list is the complete live top-level configuration registry. No numerical value used by the workflows is silently supplied by this document.

- `config/reference.yaml::schema_version`: contract version one.
- `config/reference.yaml::h`: half-thickness $h = 1$.
- `config/reference.yaml::rho`: density $\rho = 1$.
- `config/reference.yaml::lambda`: Lamé modulus $\lambda = 2$, loaded as `ReferenceConfig.lam`.
- `config/reference.yaml::mu`: shear modulus $\mu = 1$.
- `config/reference.yaml::delta`: unit cubic perturbation amplitude.
- `config/reference.yaml::epsilon_values`: ordered nonzero signed perturbations from -0.08 to 0.08 .
- `config/reference.yaml::source_radius_over_h`: $R/h = 0.5$.
- `config/reference.yaml::window_sigma_over_k0`: $\sigma/k_0 = 0.15$.
- `config/reference.yaml::window_sensitivity`: widths 0.10 and 0.20 for robustness reruns.
- `config/reference.yaml::annulus_fraction`: $f_A = 0.15$.
- `config/reference.yaml::zgv_search_kappa`: $0.20 \leq \kappa \leq 1.40$.
- `config/reference.yaml::zgv_search_omega`: $2.40 \leq \omega \leq 3.60$.
- `config/reference.yaml::eigen_residual_tolerance`: 10^{-10} .
- `config/reference.yaml::isotropic_match_tolerance`: 10^{-7} .
- `config/reference.yaml::curvature_match_tolerance`: 10^{-4} .
- `config/reference.yaml::sensitivity_match_tolerance`: 10^{-4} .
- `config/reference.yaml::phase_error_tolerance`: 0.05 rad.

Additional fixed pass/fail thresholds are part of the versioned implementation rather than the YAML file. The complete production list is:

1. The generalized matrices must be Hermitian to relative 10^{-12} ; the convergence sweep requires Hermitian defect below 10^{-12} , mass orthogonality defect below 10^{-10} , and normalized eigen residual below the configured eigen-residual tolerance. Final relative k_0, ω_0 errors must each be below the isotropic-match tolerance and the curvature error below the curvature-match tolerance.
2. Every selected eigengap must exceed ten times its stated numerical uncertainty. An angular anchor's order- p /order- $p + 4$ eigenvalue discrepancy must also be less than 0.1 times the resolved gap, angular closure MAC and principal cosine must each be at least $1 - 10^{-8}$, and a selected cluster may not touch the upper end of the twelve-mode solve. The independent branch sweep additionally requires reference-to-tracked MAC at least 0.99 and gap larger than ten times its maximum eigen residual.
3. The differentiated-mode equation and gauge residual must be at most 10^{-8} . Primary relative V and B finite-difference errors, and the last-two-step V_4 and B convergence errors, must be below the sensitivity-match tolerance. The step sweep must improve from its coarsest row and column.
4. A scaling-stage radial root must have residual no larger than

$$\max\{10 \delta g, 5 \times 10^{-10} \max(\omega/k_0, 1)\}.$$

The splitting and radial-shift log slopes must each lie within 0.1 of 1. The frequency-remainder slope must lie within 0.2 of 2. Reversing ϵ must exchange the axis and diagonal roles to angular tolerance 10^{-12} .

5. The critical stage must return eight resolved alternating roots, four minima and four saddles, whose Poincaré indices sum to zero. This workflow requirement does not claim exhaustive continuum enumeration. Candidate-grid doubling must preserve count and kind, while position, frequency, and Hessian differences obey the uncertainty bounds stated above. The negative- ϵ run must preserve angles within 2×10^{-6} and exchange minimum/saddle roles. Hessian separation, ten-times boundary separation, coarse/fine winding agreement, the fine $< \pi/2$ direction increment, and annular index closure are all mandatory.
6. A full-wave amplitude must satisfy the order-pair inequality stated in the response subsection. At each anisotropy, the minimum relative branch gap over all response surfaces must exceed ten times their maximum relative frequency uncertainty. A critical-point frequency and its independently sampled modal-node frequency must differ by no more than the node frequency uncertainty plus $64\epsilon_{\text{mach}} \max(\omega_{\text{point}}, \omega_{\text{node}}, 1)$. The three-grid complex response, rate, and phase gates in the preceding subsection are mandatory.

7. The Green workflow requires at least ten uniform-window samples. Its maximum Bessel-collapse error without fitted alignment is 0.08. The early and late slopes must be within 0.05 of $-1/2$ and -1 , respectively, and the last two grids' early and late slopes must differ by at most 0.05. The fixed-Morse relative RMS error, relative maximum error, and cancellation-region normalized RMS must not exceed 0.055, 0.17, and 0.09. The crossover-time log slope must be within 0.10 of -1 , and both production phase-discrepancy estimators must pass the configured 0.05 acceptance threshold.
8. The auxiliary response study requires strictly decreasing quadrature and interpolation errors, successive error ratios at most 0.5, finest relative quadrature error at most 0.05, and the finest accumulated interpolation phase-discrepancy estimator at most 0.05. Its baseline source/window response ratio must equal one with relative and absolute tolerances 5×10^{-13} , the robustness sweep must have nonzero range, and the widest-window boundary weight must not exceed 5×10^{-3} .
9. The silicon stress test must return eight points and pass the same noncritical-boundary and annular-index closure checks. It remains a finite-anisotropy stress test, not a weak-anisotropy proof.

At least ten samples are required in any direct log-log power-law fit; late-time beat fits additionally require at least three samples per half-open bin and at least four complete bins. Changing any fixed threshold changes the method and requires regenerating the artifact sidecars.

The committed full-profile discretizations are also fixed in the workflow modules. The isotropic stage uses GLL orders 12/16, 81 radial branch nodes, and 41 local-curvature nodes. The sensitivity stage uses order 10, 64 endpoint-free angles, constitutive differencing step 0.25, and the independent physical finite-difference step 0.0025. The weak-cubic critical-point and scaling stages use nested orders 10/14; the critical-point artifact uses $\epsilon = 0.02$, a 26×26 display grid, and Hessian step 10^{-3} . The Green stage uses nested orders 10/14, $\epsilon \in \{0.005, 0.01, 0.02, 0.04, 0.08\}$, and $t_n = 400 + 25n$ for $0 \leq n \leq 504$. The convergence stage sweeps coarse/fine order pairs 6/10, 8/12, 10/14, 12/16, 14/18, uses an independent 513×128 response reference surface, and uses sensitivity steps 0.02, 0.01, 0.005, 0.0025. The silicon stress test uses nested orders 12/16, a 40×40 display grid, and Hessian step 3×10^{-3} . These are full-profile settings; the smoke profile is a runtime check and does not replace any committed full-profile artifact.

The executable interface chain is explicit. The first seven entries below construct and track the spectral problem, the next three compute sensitivities and critical points, and the final three certify and fit the response. The six trailing constants fix the Green-workflow masks.

```
gll_nodes_weights
differentiation_matrix
build_gll_mesh
assemble_plate_matrices
assemble_wavevector_derivatives
solve_plate_modes
mass_mac
RingAnchoredSpectralEvaluator
radial_frequency_sensitivity
locate_critical_points
verify_annular_exhaustion
integrate_registered_grid_convergence
assert_phase_accuracy
fit_power_law
_EARLY_TAU_WINDOW
_FIXED_EPSILON
_MORSE_COMPARISON_START
_MORSE_COMPARISON_STOP
_MORSE_COHERENCE_THRESHOLD
_UNIFORM_TAU_MAXIMUM
```

4.6 Qualified numerical-artifact registry

The seven versioned artifacts are the sole numerical evidence consumed by the figures and manuscript. Each NPZ array is named as `artifact.key`; each value in the JSON sidecar's historically named `tolerances` namespace is `artifact.tolerances.key`. That namespace contains a mixture of configured limits and measured diagnostics; a measured value is not itself a pass threshold. The hard inequalities are the list above. The literal registry below is complete for the committed schema and sidecars. Array meanings are grouped before it to make the mapping operational rather than decorative.

Isotropic baseline. The isotropic artifact stores the sampled symmetric branch, exact ZGV coordinates and curvature, its local quadratic comparison, and the selected thickness mode and squared-displacement proxy. Its sidecar binds exact/spectral mismatch, nested-order frequency uncertainty, and eigengap.

Angular perturbation. The sensitivity artifact stores $V(\theta)$, $B(\theta)$, their finite-difference checks, cubic harmonic coefficients, and the physical signed shift ϵV_4 . Its sidecar binds eigenpair, symmetry, harmonic-leakage, and relative derivative errors.

Critical points and scaling. The critical-point artifact stores the eight full-wave roots, Cartesian Hessian eigenvalues and Poincaré indices, perturbative predictions, and the two plotted Cartesian surfaces. The scaling artifact stores signed ϵ frequency and radial shifts, their compensated remainders, role reversal, and measured log slopes.

Response, convergence, and stress test. The response artifact stores three complex responses: direct, uniform-normal-form, and exact-Morse. It also stores masks, spectra, slopes, critical frequencies, and phase errors. The convergence artifact holds the polynomial, tracking, response-grid, finite-difference, and source/window sweeps. The silicon artifact is a finite-anisotropy stress test only. It is not evidence for the weak-anisotropy theorem.

```
isotropic_zgv.kappa
isotropic_zgv.omega_symmetric
isotropic_zgv.branch_labels
isotropic_zgv.kappa0
isotropic_zgv.omega0
isotropic_zgv.curvature_a
isotropic_zgv.local_q
isotropic_zgv.local_omega
isotropic_zgv.local_quadratic
isotropic_zgv.mode_z
isotropic_zgv.mode_u
isotropic_zgv.mode_squared_displacement
isotropic_zgv.tolerances.configured_isotropic_match
isotropic_zgv.tolerances.maximum_frequency_uncertainty
isotropic_zgv.tolerances.minimum_relative_eigengap
isotropic_zgv.tolerances.relative_isotropic_frequency

angular_sensitivity.theta
angular_sensitivity.V
angular_sensitivity.B
angular_sensitivity.V_reconstruction
angular_sensitivity.harmonic_order
angular_sensitivity.harmonic_amplitude
angular_sensitivity.V0
angular_sensitivity.V4
angular_sensitivity.V8
angular_sensitivity.epsilon
angular_sensitivity.delta_c
angular_sensitivity.physical_V4_shift
angular_sensitivity.V_fd
angular_sensitivity.B_fd
angular_sensitivity.tolerances.configured_relative_sensitivity
angular_sensitivity.tolerances.maximum_eigen_residual
angular_sensitivity.tolerances.minimum_relative_eigengap
angular_sensitivity.tolerances.mirror_defect
angular_sensitivity.tolerances.non_cubic_leakage
angular_sensitivity.tolerances.periodicity_defect
angular_sensitivity.tolerances.relative_B_error
angular_sensitivity.tolerances.relative_V_error

critical_points.kx
critical_points.ky
critical_points.kappa
critical_points.theta
critical_points.omega
critical_points.hessian_eigenvalues
critical_points.morse_index
critical_points.kind
critical_points.kx_pred
critical_points.ky_pred
critical_points.omega_pred
critical_points.gradient_residual
critical_points.kx_grid
critical_points.ky_grid
critical_points.omega_iso_grid
critical_points.omega_aniso_grid
```

critical_points.tolerances.epsilon
 critical_points.tolerances.maximum_gradient_residual
 critical_points.tolerances.maximum_grid_frequency_uncertainty
 critical_points.tolerances.minimum_hessian_to_uncertainty
 critical_points.tolerances.negative_candidate_grid_frequency_difference
 critical_points.tolerances.negative_candidate_grid_hessian_difference
 critical_points.tolerances.negative_candidate_grid_position_difference
 critical_points.tolerances.negative_maximum_boundary_gradient_uncertainty
 critical_points.tolerances.negative_minimum_boundary_gradient
 critical_points.tolerances.positive_candidate_grid_frequency_difference
 critical_points.tolerances.positive_candidate_grid_hessian_difference
 critical_points.tolerances.positive_candidate_grid_position_difference
 critical_points.tolerances.positive_maximum_boundary_gradient_uncertainty
 critical_points.tolerances.positive_minimum_boundary_gradient

 perturbation_scaling.epsilon
 perturbation_scaling.delta_omega_full
 perturbation_scaling.delta_omega_pred
 perturbation_scaling.q_min_full
 perturbation_scaling.q_min_pred
 perturbation_scaling.q_saddle_full
 perturbation_scaling.q_saddle_pred
 perturbation_scaling.omega_min_error
 perturbation_scaling.omega_saddle_error
 perturbation_scaling.compensated_splitting
 perturbation_scaling.compensated_q_min
 perturbation_scaling.compensated_q_saddle
 perturbation_scaling.compensated_frequency_error
 perturbation_scaling.role_reversal_theta_min
 perturbation_scaling.role_reversal_theta_saddle
 perturbation_scaling.slope_splitting
 perturbation_scaling.slope_radial
 perturbation_scaling.slope_remainder
 perturbation_scaling.tolerances.isotropic_axis_radius_bias
 perturbation_scaling.tolerances.isotropic_diagonal_radius_bias
 perturbation_scaling.tolerances.maximum_frequency_uncertainty
 perturbation_scaling.tolerances.maximum_gradient_residual
 perturbation_scaling.tolerances.slope_radial_error
 perturbation_scaling.tolerances.slope_remainder_error
 perturbation_scaling.tolerances.slope_splitting_error

 green_crossover.epsilon
 green_crossover.time
 green_crossover.G_full
 green_crossover.G_normal_form
 green_crossover.G_morse
 green_crossover.tau
 green_crossover.J0
 green_crossover.scaled_response
 green_crossover.envelope
 green_crossover.rms_envelope
 green_crossover.local_slope
 green_crossover.fit_window_early
 green_crossover.fit_window_late
 green_crossover.slope_early
 green_crossover.slope_late
 green_crossover.crossover_time
 green_crossover.spectrum_omega
 green_crossover.spectrum
 green_crossover.omega_min
 green_crossover.omega_saddle
 green_crossover.phase_error
 green_crossover.tolerances.bessel_collapse_error_by_epsilon
 green_crossover.tolerances.configured_phase_error
 green_crossover.tolerances.fixed_epsilon_cancellation_fraction
 green_crossover.tolerances.fixed_epsilon_cancellation_region_normalized_rms
 green_crossover.tolerances.fixed_epsilon_morse_relative_max_error
 green_crossover.tolerances.fixed_epsilon_morse_relative_rms_error
 green_crossover.tolerances.maximum_bessel_collapse_error
 green_crossover.tolerances.maximum_phase_error
 green_crossover.tolerances.maximum_relative_frequency_uncertainty_by_epsilon
 green_crossover.tolerances.measured_crossover_log_slope

```

green_crossover.tolerances.measured_early_slope_error
green_crossover.tolerances.measured_late_slope_error
green_crossover.tolerances.minimum_anisotropic_branch_eigengap_by_epsilon
green_crossover.tolerances.registered_accumulated_phase_errors
green_crossover.tolerances.registered_complex_response_errors
green_crossover.tolerances.registered_early_slope_by_grid
green_crossover.tolerances.registered_late_slope_by_grid
green_crossover.tolerances.registered_nested_frequency_errors

convergence.polynomial_order
convergence.omega0_error
convergence.kappa0_error
convergence.curvature_error
convergence.eigen_residual
convergence.hermitian_residual
convergence.mass_orthogonality
convergence.eigengap
convergence.angular_resolution
convergence.radial_resolution
convergence.quadrature_error
convergence.interpolation_error
convergence.phase_error
convergence.sensitivity_step
convergence.V4_fd_error
convergence.B_fd_error
convergence.tracking_kappa
convergence.tracking_mac
convergence.tracking_gap
convergence.source_width
convergence.window_width
convergence.response_sensitivity
convergence.tolerances.configured_curvature_match
convergence.tolerances.configured_eigen_residual
convergence.tolerances.configured_isotropic_match
convergence.tolerances.configured_phase_error
convergence.tolerances.final_B_fd_relative_error
convergence.tolerances.final_V4_fd_relative_error
convergence.tolerances.final_relative_curvature_error
convergence.tolerances.final_relative_kappa0_error
convergence.tolerances.final_relative_omega0_error
convergence.tolerances.finest_accumulated_phase_error
convergence.tolerances.finest_interpolation_frequency_error
convergence.tolerances.finest_quadrature_relative_error
convergence.tolerances.maximum_eigen_residual
convergence.tolerances.maximum_hermitian_residual
convergence.tolerances.maximum_mass_orthogonality
convergence.tolerances.minimum_eigengap
convergence.tolerances.minimum_tracking_gap
convergence.tolerances.minimum_tracking_mac

silicon_stress_test.kx_grid
silicon_stress_test.ky_grid
silicon_stress_test.omega_grid
silicon_stress_test.kx
silicon_stress_test.ky
silicon_stress_test.omega
silicon_stress_test.hessian_eigenvalues
silicon_stress_test.kind
silicon_stress_test.tracking_gap
silicon_stress_test.gradient_residual
silicon_stress_test.tolerances.maximum_boundary_gradient_uncertainty
silicon_stress_test.tolerances.maximum_gradient_residual
silicon_stress_test.tolerances.minimum_boundary_gradient
silicon_stress_test.tolerances.minimum_tracking_gap

```

4.7 Reconstruction boundary

An independent reader can reconstruct the calculation from the weak-form matrices, the qualified configuration, and the listed interfaces. The numerical evidence is limited to the tracked local branch inside the registered annulus and spectral window. It does not claim completeness of the full plate spectrum outside that window, and the finite-silicon run does not extend the weak-anisotropy proof beyond its perturbative assumptions.

5 Generated convergence records and robustness figures

This section collects the generated records that accompany the executable numerical specification in Supplementary Section 4. The tables are regenerated from qualified artifacts and the registered configuration; the figures read only the same artifacts and export the plotted arrays to `data/source_data/supplementary`. They are evidence for discretization, phase, branch-tracking, differentiation, source-window, and finite-anisotropy robustness within the declared ranges, not replacements for the hypotheses of the perturbation and stationary-phase theorems.

Table 1: Independent isotropic one-element and two-element convergence record. The errors are relative to the high-precision Rayleigh–Lamb ZGV root; the residual and eigengap use the definitions in Supplementary Section 4.

discretization	p	elements	$ \delta\kappa_0 /\kappa_0$	$ \delta\Omega_0 /\Omega_0$	$ \delta a /a$	eigen residual	minimum eigengap
one element	28	1	1.642797e-14	2.144017e-12	8.610456e-11	4.685472e-12	0.22289169
two elements	28	2	1.451459e-12	5.332013e-13	1.225559e-09	8.997835e-12	0.22289169

Table 2: Registered reference parameters, numerical tolerances, and the material record used only for the silicon stress test. Dimensionless entries are read from the registered YAML configuration; the silicon constants retain their source identifier in the generated table.

parameter	value	provenance
h	1	reference configuration
ρ	1	reference configuration
λ	2	reference configuration
μ	1	reference configuration
δ	1	reference configuration
R/h	0.5	reference configuration
σ/k_0	0.15	reference configuration
annulus fraction	0.15	reference configuration
eigen residual tolerance	1.000000e-10	reference configuration
isotropic match tolerance	1.000000e-07	reference configuration
curvature match tolerance	1.000000e-04	reference configuration
sensitivity match tolerance	1.000000e-04	reference configuration
phase error tolerance	0.05	reference configuration
C_{11} (GPa)	165.7	doi:10.1107/S1600577514004962
C_{12} (GPa)	63.9	doi:10.1107/S1600577514004962
C_{44} (GPa)	79.6	doi:10.1107/S1600577514004962

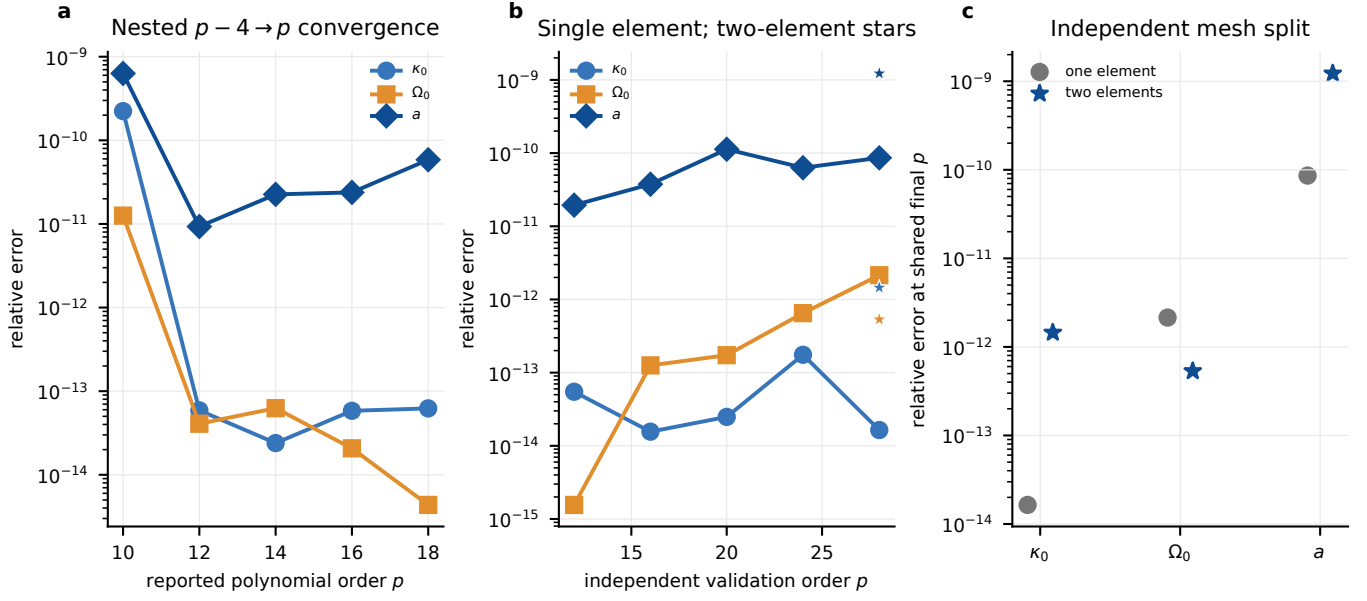


Figure 1: Polynomial-order and assembly cross-checks for the selected isotropic branch. **a**, Solid marker–lines give relative errors in the exact Rayleigh–Lamb reference κ_0 (blue circles), Ω_0 (orange squares), and curvature a (dark-blue diamonds) on logarithmic axes. Each reported fine GLL order p pairs a tracked evaluator at $p - 4$ with its independent diagnostic at p ; the machine-precision plateaux need not be monotone. **b**, The same colours and markers show an independently reconstructed one-element spectrum, while the coloured stars at the final order report the independent two-element result. **c**, At the shared final order, grey circles and dark-blue stars compare the one- and two-element errors for all three quantities. The declared gates are relative κ_0 and Ω_0 errors below 10^{-7} , relative curvature error below 10^{-4} , eigen residual below 10^{-10} , and relative eigengap greater than ten times the larger of that residual and machine precision. These are local discretization and mesh-split diagnostics, not a proof for the complete spectrum. Source data: `data/source_data/supplementary`.

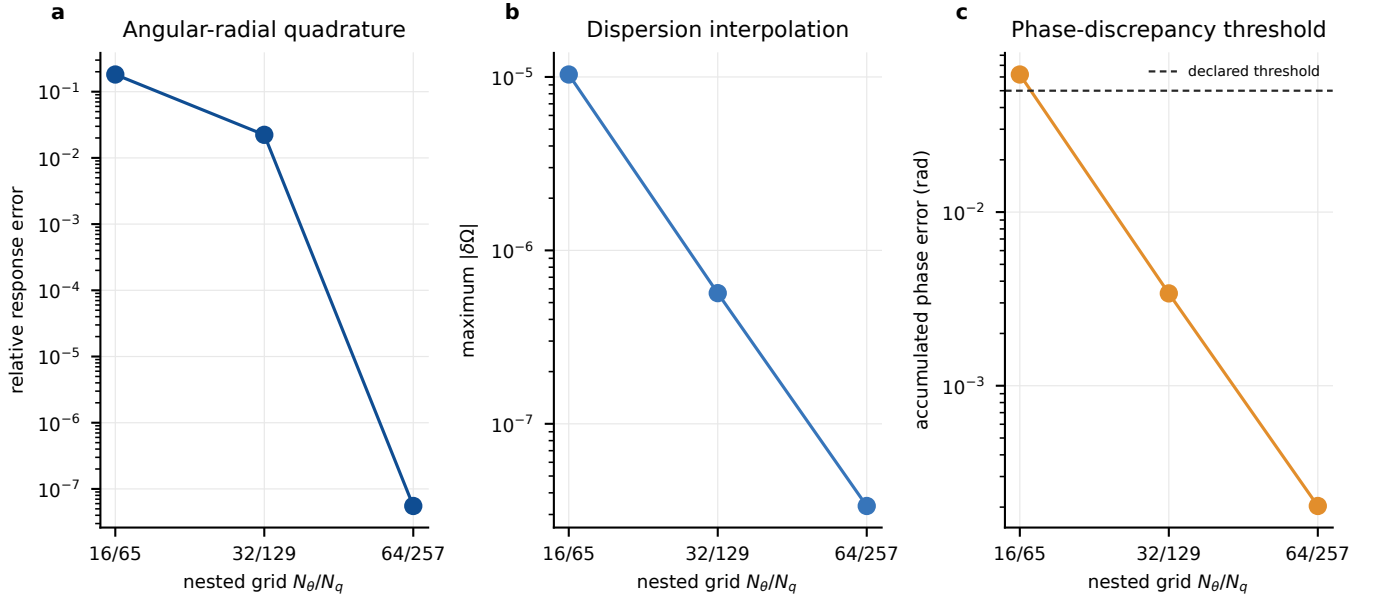


Figure 2: Quadrature, interpolation, and phase-discrepancy convergence for the declared response window. All panels use the nested angular-radial grids $(N_\theta, N_q) = (16, 65), (32, 129), (64, 257)$ and a $(128, 513)$ reference surface; solid curves with circular markers and logarithmic ordinates show directly computed diagnostics. **a**, Blue circles give the RMS norm of the coarse-minus-reference demodulated complex full-wave response divided by the reference-response RMS over the declared time samples. **b**, Blue circles give the maximum absolute frequency error after periodic-angular and cubic-radial interpolation of each coarse surface onto the reference grid. **c**, Orange circles give the accumulated inter-resolution phase-discrepancy estimator at the declared maximum time, including interpolation and coarse/reference nested-order frequency differences; the black short-dashed line is the declared acceptance threshold of 0.05 rad. The finest quadrature error is required to be at most 0.05, and successive quadrature and interpolation ratios at most one half. These empirical estimators test consistency across computed resolutions; they do not bound the unknown continuum error or imply infinite-time phase accuracy. Source data: `data/source_data/supplementary`.

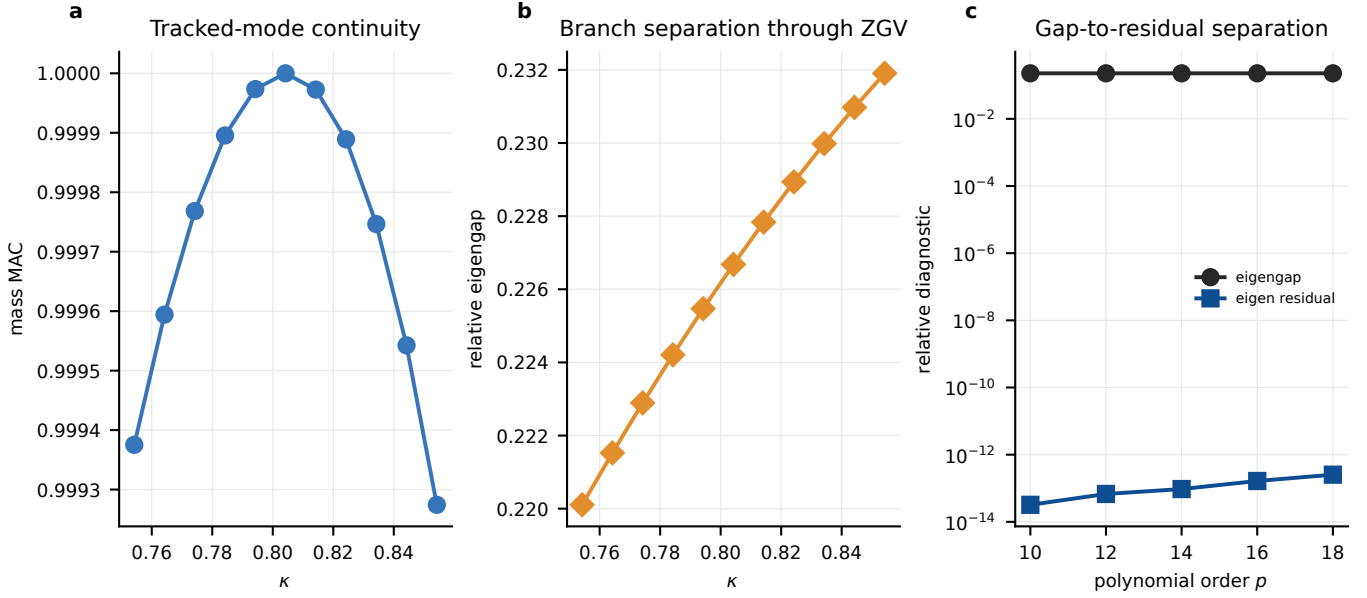


Figure 3: Local branch-tracking and simple-eigenvalue diagnostics. **a**, The solid blue curve with circles is the mass-weighted modal assurance criterion relative to the κ_0 anchor over the declared local wavenumber window; the acceptance threshold is 0.99. **b**, The solid orange curve with diamonds is the relative eigengap of the same tracked branch over that window. **c**, On a logarithmic ordinate, black circles and a solid line report the eigengap and dark-blue squares and a solid line report the normalized eigen residual over the polynomial-order sweep. Acceptance requires residual below 10^{-10} and gap greater than ten times the larger of residual and machine precision. Together these computed quantities support branch continuity and separation only near the selected ZGV point; they neither identify the branch globally nor certify unrelated modes. Source data: `data/source_data/supplementary`.

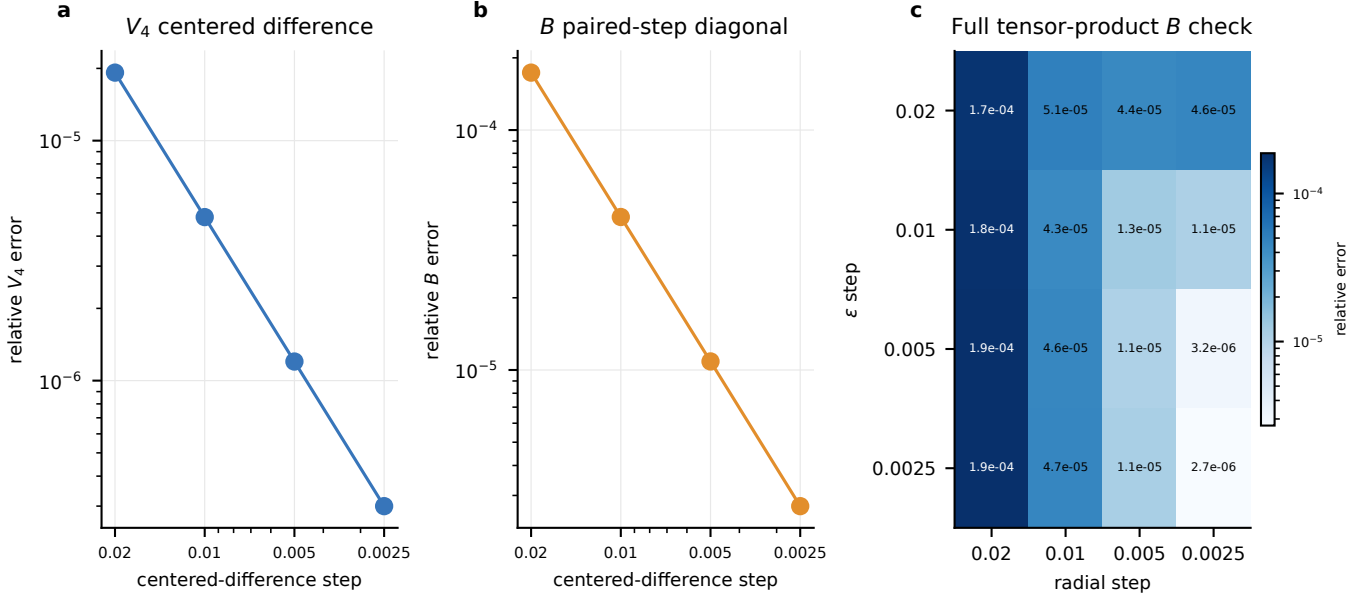


Figure 4: Independent finite-difference closure of the analytic generalized-eigenvalue sensitivities. The horizontal step axes in panels **a** and **b** are reversed so refinement runs from left to right. **a**, Blue circles and a solid line show the centered-difference relative error in V_4 versus $\Delta\epsilon$. **b**, Orange circles and a solid line show the paired-step diagonal error in B , defined as the Euclidean relative error of the two-component vector at $\theta = 0$ and $\pi/4$ when $\Delta\epsilon = \Delta q$. **c**, The logarithmically coloured tensor-product heatmap reports the same vector error for every pair $(\Delta\epsilon, \Delta q)$; darker cells denote larger error and printed annotations give the numerical values. The sensitivity tolerance is 10^{-4} : the final two V_4 steps and the finest two-by-two B block satisfy it. These full-wave centered differences are independent of the analytic sensitivity formula, although they use the same full-wave eigensolver; they are not fitted data. Source data: [data/source_data/supplementary](#).

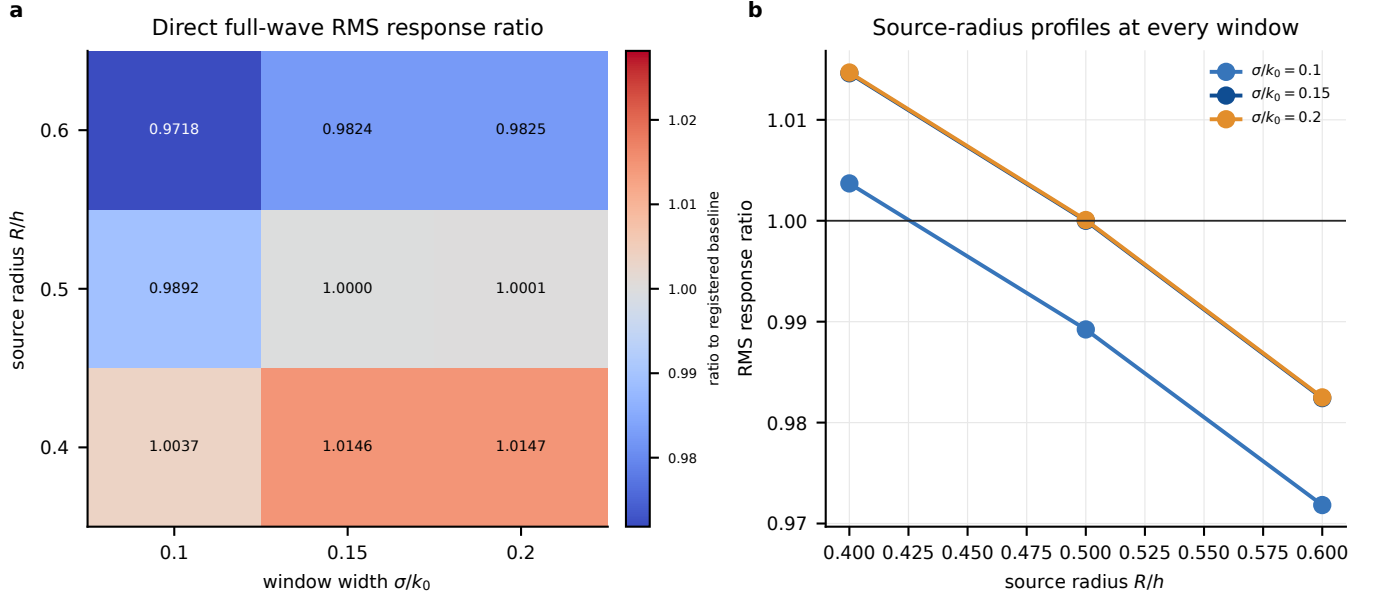


Figure 5: Sensitivity of the computed response norm to the declared source and spectral window class. Each ratio uses the RMS of the demodulated complex full-wave response over the declared time samples at $\varepsilon = 0.02$, normalized by the baseline $(R/h, \sigma/k_0) = (0.5, 0.15)$. **a**, The annotated heatmap covers three source radii and three super-Gaussian widths; the diverging colour scale is centered on the normalized baseline value one. **b**, Solid curves with circular markers show the same ratios against R/h , with colour distinguishing σ/k_0 ; the black horizontal line is the declared baseline. The separately computed widest-window boundary weight is required to remain below the truncation gate 5×10^{-3} . This sweep supports response-norm robustness within the displayed source–window class only. It does not establish source independence and is not evidence for either the crossover exponents or the Morse theorem. Source data: `data/source_data/supplementary`.

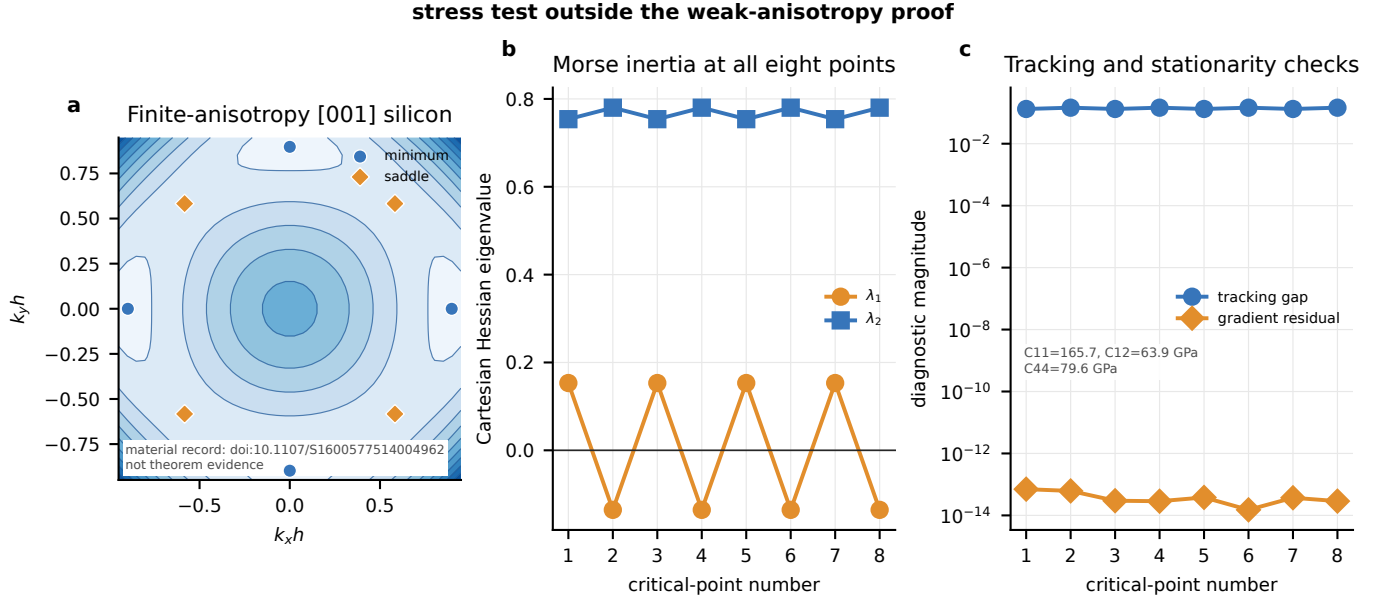


Figure 6: Finite-anisotropy silicon stress test outside the weak-anisotropy theorem, for a [001] plate with $C_{11} = 165.7$, $C_{12} = 63.9$, and $C_{44} = 79.6$ GPa (material source doi:10.1107/S1600577514004962). **a**, Filled blue shading and blue contour lines show the full-wave local-frequency surface normalized as $\Omega = \omega h / \sqrt{C_{44}/\rho}$; blue circles and orange diamonds mark four refined minima and four saddles within the declared annulus. The shading is not a separate quantitative colour scale. **b**, Points are ordered by stored index; orange circles and a solid line show the Cartesian Hessian eigenvalue λ_1 , blue squares and a solid line show λ_2 , and the black line marks zero. Here marker identity denotes eigenvalue rather than Morse class: both values are positive at minima and have opposite signs at saddles. **c**, On a logarithmic axis, blue circles and a solid line give the relative tracking gap, while orange diamonds and a solid line give the stationarity residual $\|\nabla_{\mathbf{k}}\Omega\|$; this panel does not display the separately evaluated annular-boundary consistency check. This is a finite-anisotropy stress test of the numerical search and local classification only, not evidence for the weak-anisotropy theorem. Source data: data/source_data/supplementary.